Chapter 1
Introduction

(Lecture of 6 class hours)

Outline

- *What is a Computer Network?*
- What can we do with Computer Networks?
- Categories of Computer Networks
- Network architecture and protocols
- Reference Models
- Example Networks
- Network Standardization

What is a Computer Network?

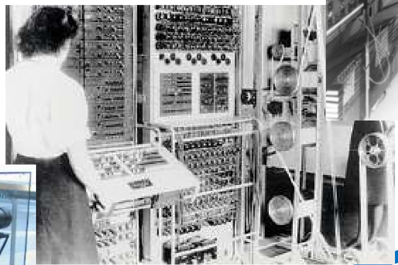
a group of interconnected computers

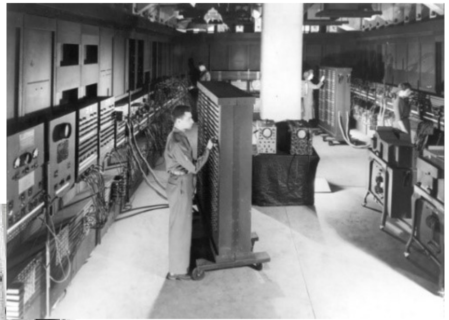
This involves...

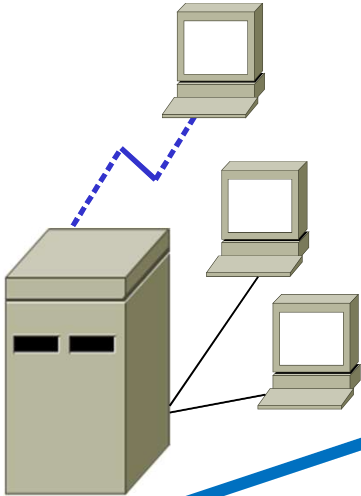
- What can be connected?
  - Computers, PADs, Smart phones, Home appliances, Sensors, *Anything?*
- How to connect?
  - 直接相连 or 需要中转（中转设备：HUB or Router）
- What kind of links?
  - 有线（Wired） or 无线（wireless）
- How far can be reached?
  - 相邻几个房间（LAN），一个省/国家内（WAN）

    or 全球（Internet/因特网）

Review of History(1)







1942: ABC
阿塔纳索夫-贝瑞计算机
不可编程

1943-1945: Colossus
巨人计算机
不能存储程序

1946: ENIAC (by Mauchly+ Eckert)
第一台通用电子计算机

1960s:
联机系统
（主机带终端）

终端：有键盘和显示器，没有处理能力，不能独立工作

1969: APARNet
计算机网络

Review of History(2)

	1970s	1980s	1990s	2000s	2010--
网络应用 Applications	E-mail BBS 文件下载 (文本传输)	WWW (1989) (超媒体)	电子商务 即时消息 (ICQ、QQ) BLOG	VoIP IPTV P2P 社交网络(Facebook) 自媒体(Twitter, 微博) 音乐视频分享 (Youtube) 团购网 云计算	移动社交 (微信) 互联网+
网络技术 Networks	数百台主机 连接到 ARPANet; TCP/IP协议 开发	NSFNet→ Internet, TCP/IP支持 异构网, LAN	WiFi	传感网 物联网 (IoT)	共享经济 (共享单车/ 嘀嘀快车) 移动支付 (线下支付/ 外卖）
主机硬件 Hardware	大型主机	PC 掌上电脑	智能手机 (IBM,1993) 触屏式平板电脑		大数据 短视频
带宽 Bandwidth	100kbps	1~10Mbps	10~100Mbps	1~10Gbps	100Gbps

Concepts may be Confused

- Communication vs. Computer
  - Communication: process of transferring information from one entity to another.(*wikipeida*)
  - Computer: machine helping human beings to process information.
- Communication Network vs. Computer Network
  - Communication Network(Telecommunication Network): a network of telecommunications links and nodes to transfer messages
  - Computer Network: a group of interconnected computers
- Distributed Systems vs. Computer Networks

Distributed System(分布式系统)

- A collection of independent computers appears to its users as a single coherent system
- A single model or paradigm that it presents to the users
  - Transparency(透明：不可见)
  - "The illusion that you have something as simple as a uniprocessor system."
- A well-known example：the World Wide Web,
  - everything looks like a document (Web page)

What is a Computer Network

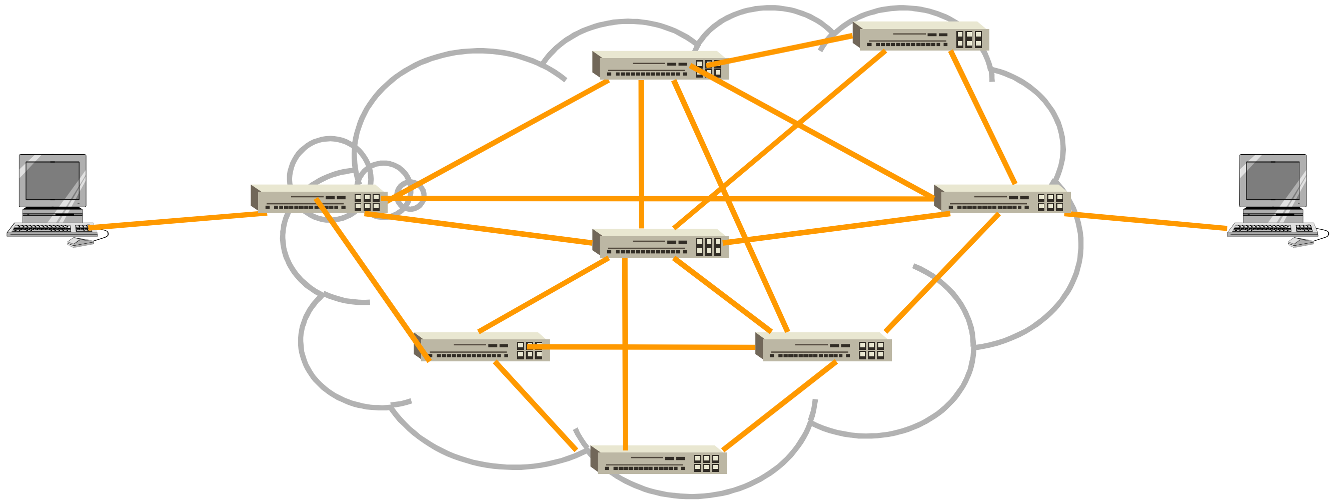
- Computer Network
  - A collection of autonomous computers interconnected by a single technology
  - Users are exposed to the actual machines
  - If the machines have different hardware and different operating systems, that is fully visible to the users
- However
  - many problems in common
  - Some networks or parts of them (e.g., name services) are also distributed systems
  - every distributed system relies on services provided by a computer network

Computer Network is not

- Internet（因特网）
  - A network of networks
  - The unique global network
- WWW（万维网）
  - A distributed system (or an application) running on top of Internet

Hardware Components of Computer Network

- Computers/Hosts(主机)/End Systems
- Communication Links (wired or wireless)
- Switches/Routers (Nodes, 节点)



Cool end systems(端系统)

Smart speaker +
IP picture frame

Internet
refrigerator

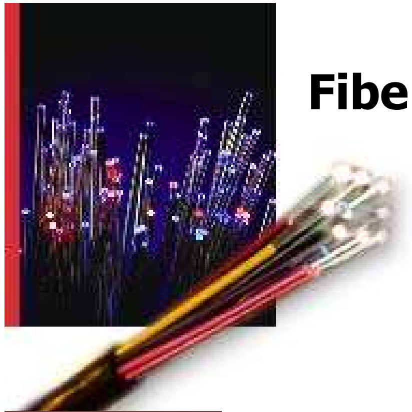
Web-enabled toaster +
weather forecaster

Wearable device:
Smart bracelet

Internet phones

Nodes and Communication Links

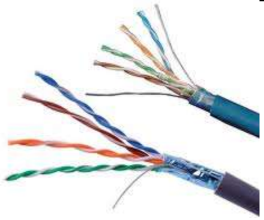
Links



Fiber

Coax

Twisted Pair



Network Interface

Ethernet NIC

Wireless NIC

Router/Switch

Router

LAN Switch

[Kurose]

Tri-networks integration: 三网融合 merging of 3 types of networks

- Computer Networks
- Communication Networks
- Cable TV Networks
- The 4th type: electrical network (Power-line networking, 电力网)

Outline

- What is a Computer Network?
- *What can we do with Computer Networks?*
  - *Business applications*
  - *Home applications*
  - *Mobile network users*
  - *Social Issues*
- Categories of Computer Networks
- Network architecture and protocols
- Reference Models
- Example Networks
- Network Standardization

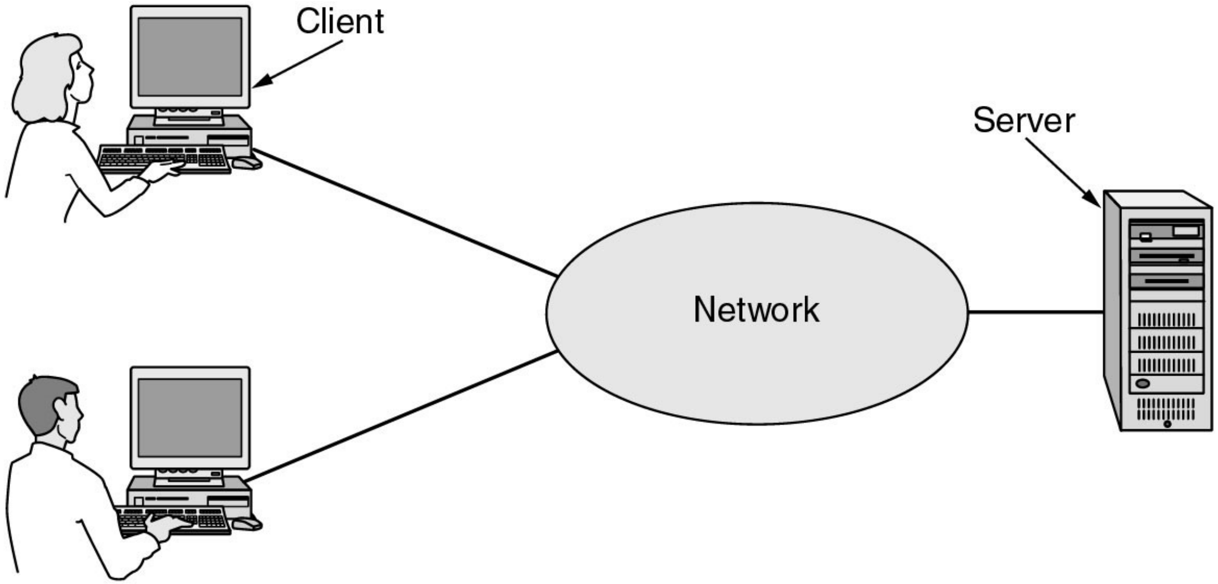
中国互联网络发展状况统计报告

Business Applications

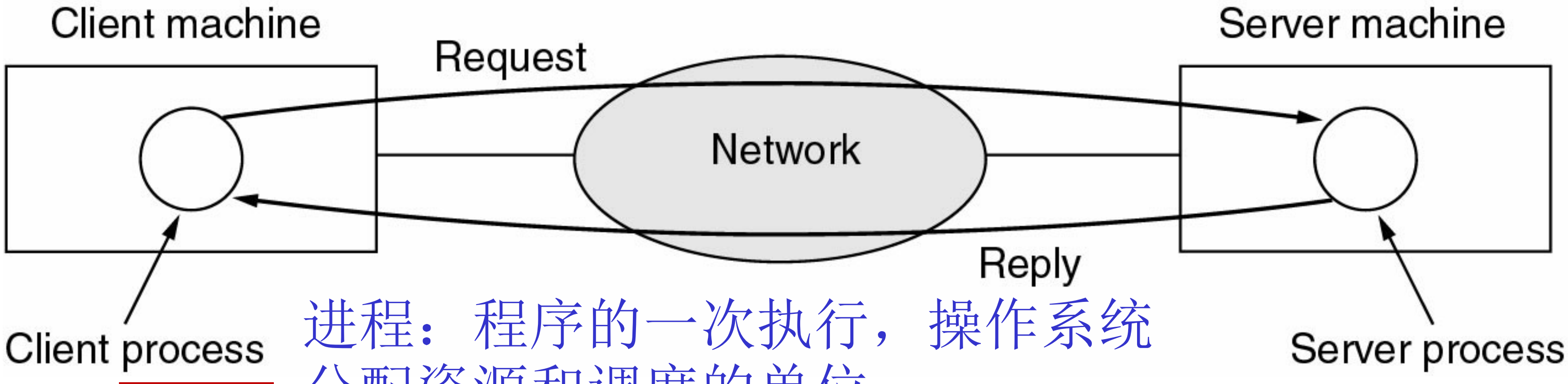
- Resource sharing
  - Printers, scanners, database, documents…
- Communication medium among employees
  - Email, VoIP, videoconferencing…
- Electronic business
  - Doing business electronically with other companies
- Electronic commerce
  - Airlines, bookstores, shopping carts,…
- ……

Client-Server Model

- A network with two clients and one server



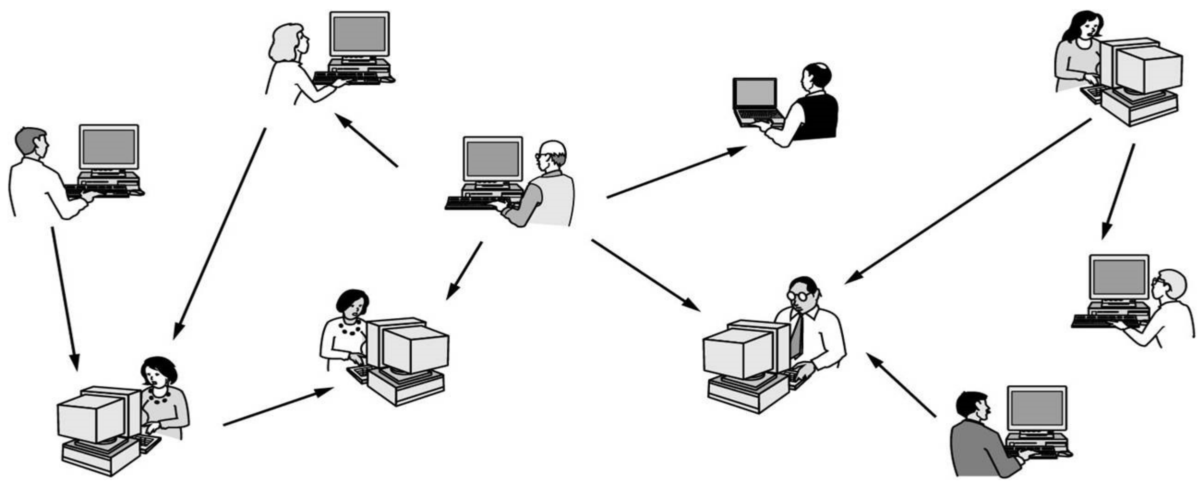
存储资源、
提供服务



进程：程序的一次执行，操作系统
分配资源和调度的单位

Home Applications

- Access to remote information
  - Web browsing
- Person-to-person communication
  - Instant messaging (QQ), social networks(WeChat)
- Interactive entertainment
  - Game playing
- Electronic commerce
- Ubiquitous computing (普适计算)



peer-to-peer(P2P) model

Mobile Network Users

- Combinations of wireless networks and mobile computing.

Wireless	Mobile	Applications
No	No	Desktop computers in offices
No	Yes	A notebook computer used in a hotel room
Yes	No	Networks in older, unwired buildings
Yes	Yes	Portable office; PDA for store inventory

Mobile Network Users

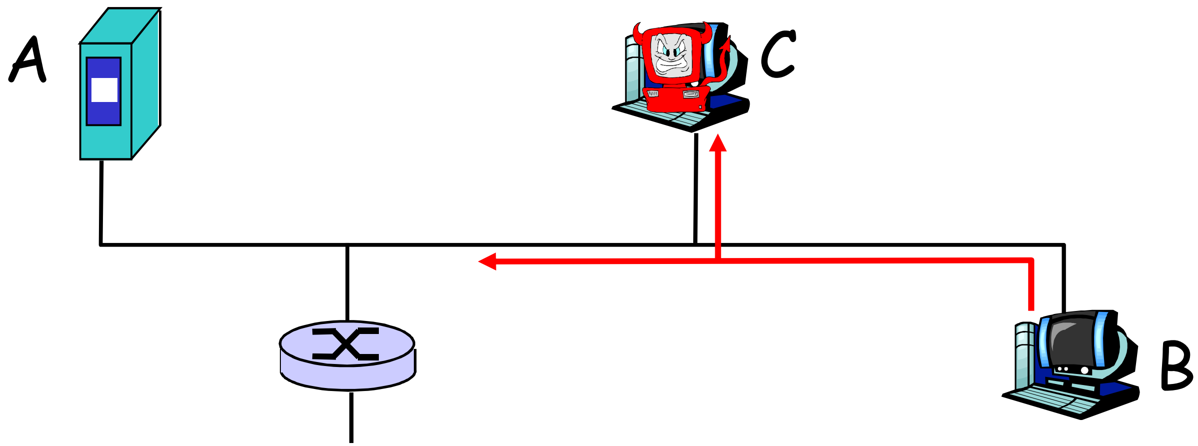
- Portable office
- Vehicle communication: trucks, taxis, …, keeping in contact with each other and with home office
- M-commerce(移动商务)
- RFID
- Sensor networks
- Smart watches
- …..

Usage of RFID：智能餐盘

- 在餐盘底部植入RFID射频芯片
- 餐盘放到结算台的指定区域，可自动识别出价格
- 读写频率：HF13.56MHz
- 餐盘材质： 密胺
- 耐受温度：260℃
- 读写次数：>10万次

Social Issues

- Politics, religion, sex
- Eavesdropping(窃听)
- Credit card
- Privacy
- Unwanted communication (Spam, …)



Outline

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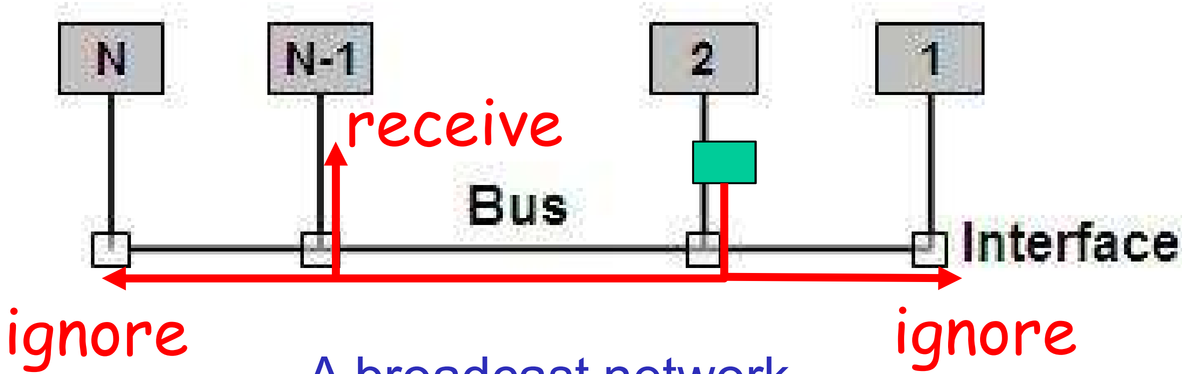
Category of Computer Networks

- Computer Networks can be categorized in a number of ways
  - Transmission technology
    - Broadcast networks vs. Point-to-point networks
  - Network scale
    - PAN, LAN, MAN, WAN
  - Position in the interconnected networks
    - Access networks vs. Core networks

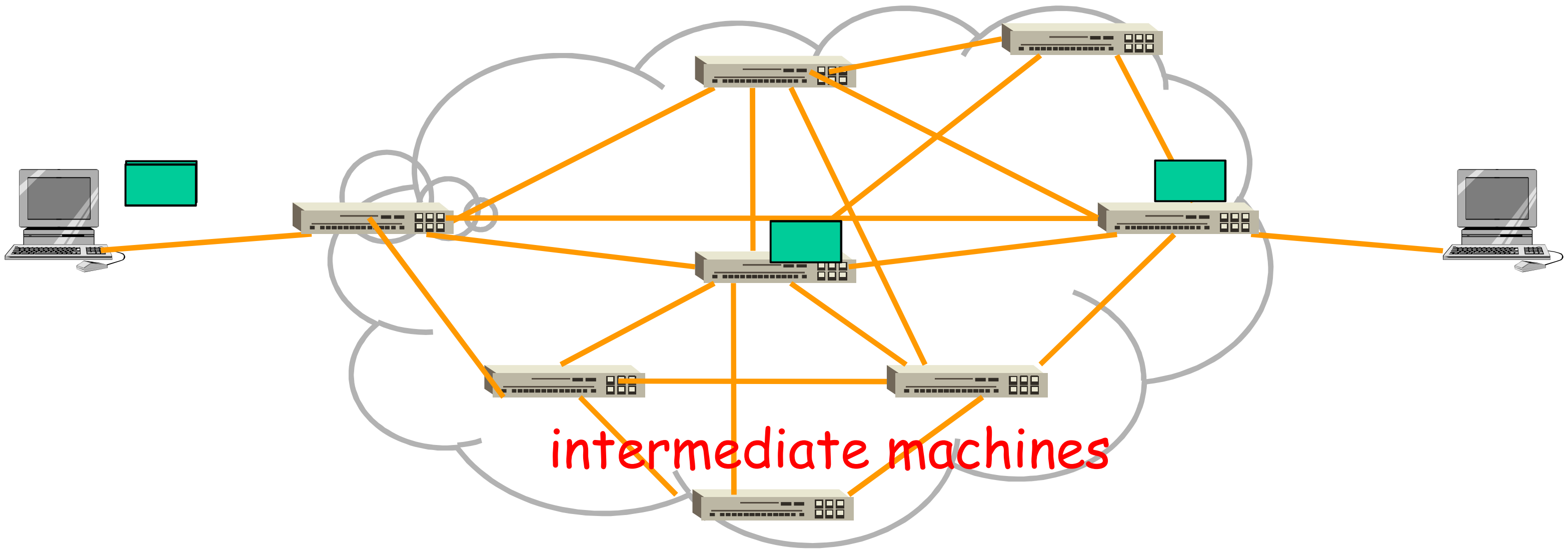
Classification on transmission technology

- **Broadcast networks (广播网络)**
  - Sending a packet to all destinations, each machine checks the address field
    - If the packet is intended for the receiving machine, that machine processes the packet
    - If the packet is intended for some other machine, it is just ignored.
  - Broadcasting vs. Multicasting(组播/多播)
  - LANs
- **Point-to-point networks (点到点网络)**
  - To go from the source to the destination, a packet may visit one or more intermediate machines
  - WANs

Broadcast vs. Point-to-point



A broadcast network



A point-to-point network

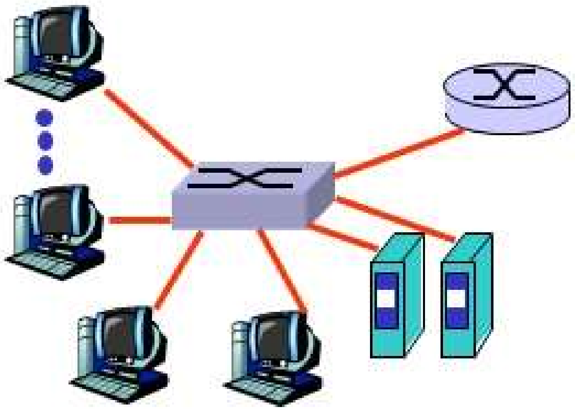
Classification by scale

Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network 个域网
10 m	Room	Local area network 局域网
100 m	Building	
1 km	Campus	
10 km	City	Metropolitan area network 城域网
100 km	Country	Wide area network 广域网
1000 km	Continent	
10,000 km	Planet	The Internet

Except for scales, the technology used are different!

LAN(局域网)

- Connecting hosts to edge router
- Sharing ONE communication link



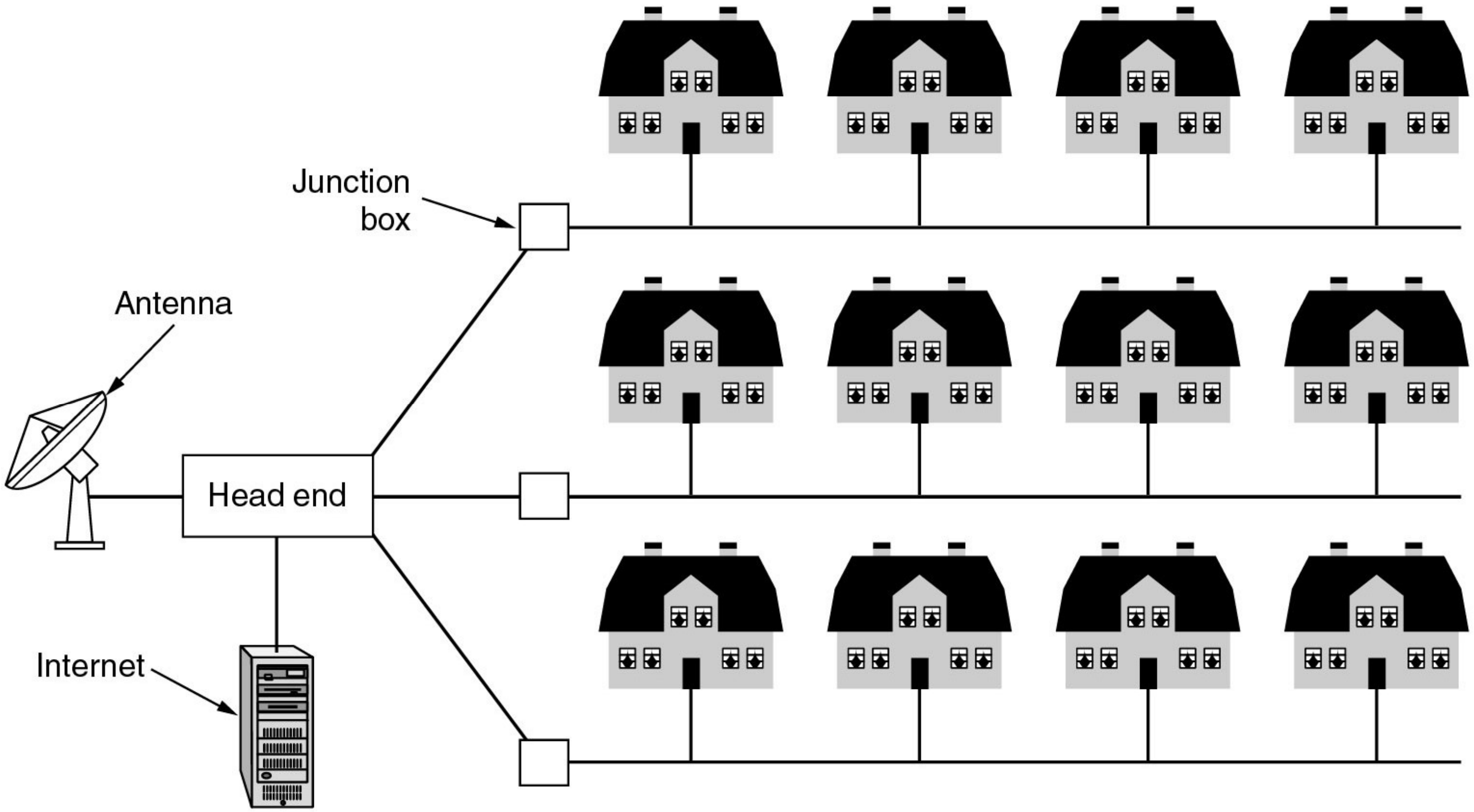
Ethernet

WLAN(WiFi)

[Kurose]

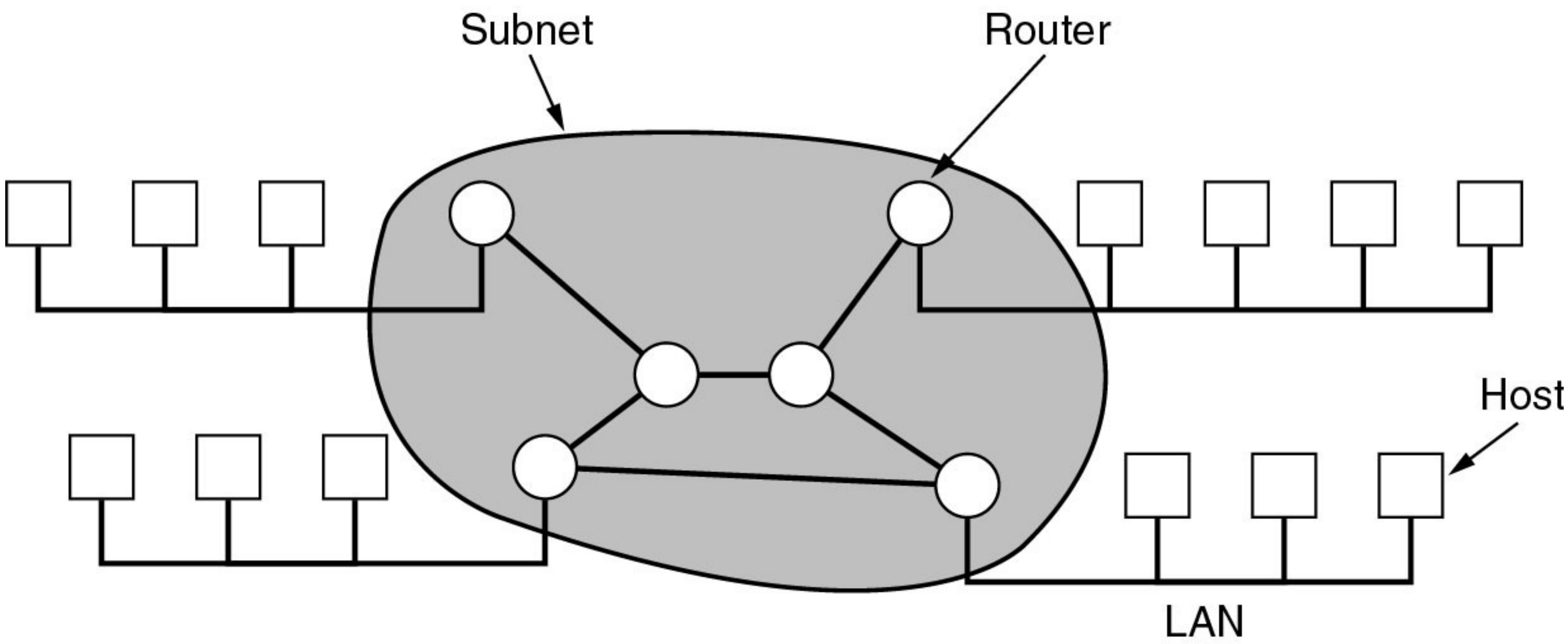
Metropolitan Area Networks(城域网)

- A metropolitan area network based on cable TV.
- Topology: Tree



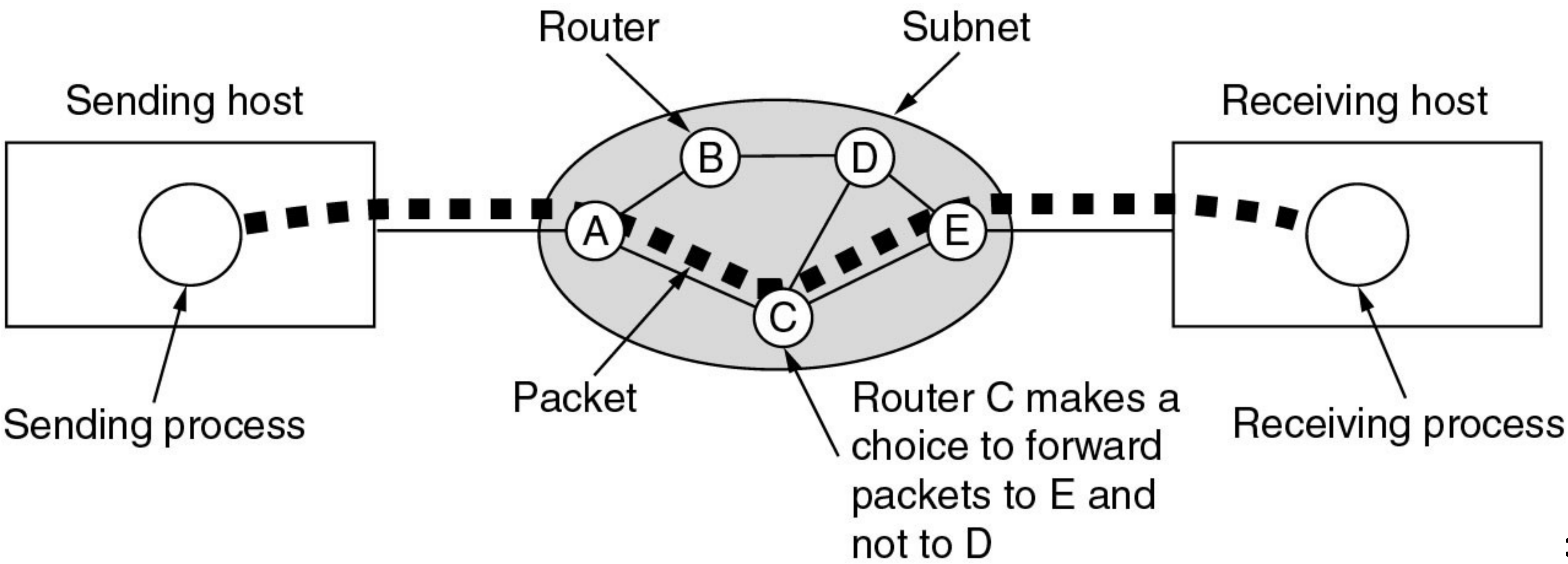
WAN(广域网): LAN and Subnet

- WAN: telecommunication network that covers a broad area
- Providing connections from a LAN to the Internet
- Topology: Mesh（网孔型）
- Communication oriented



WAN: Communication subnet

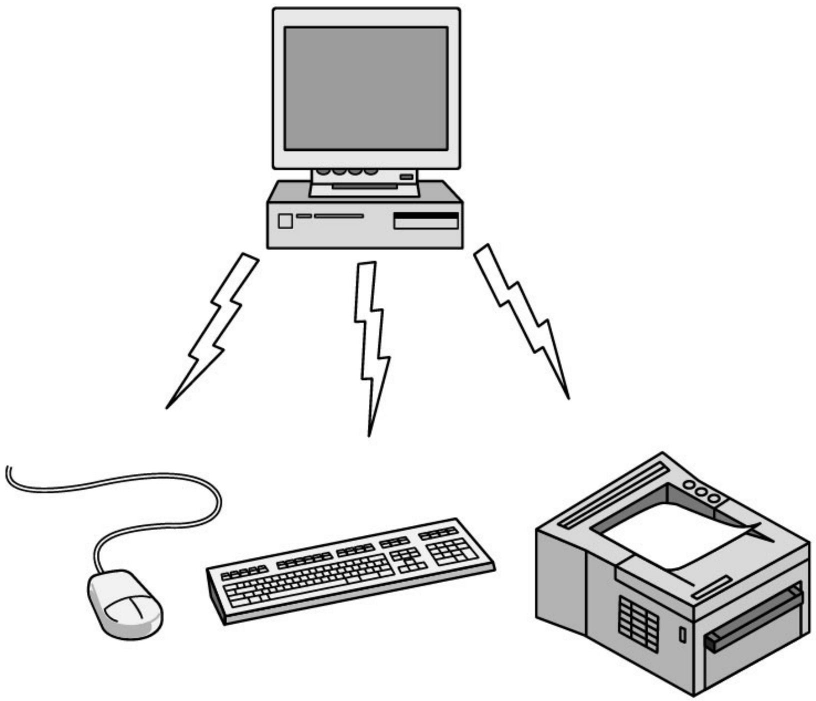
- **Switching elements (交换设备：Router)**
  - **Store-and-forward (存储转发)** (packet-switched, 分组交换)
  - Routing decisions: Routing algorithm
- **Message Path**
  - Single-path vs. Multi-path
  - Best route must be used
  - Routing decisions are made locally



Wireless networks: categories

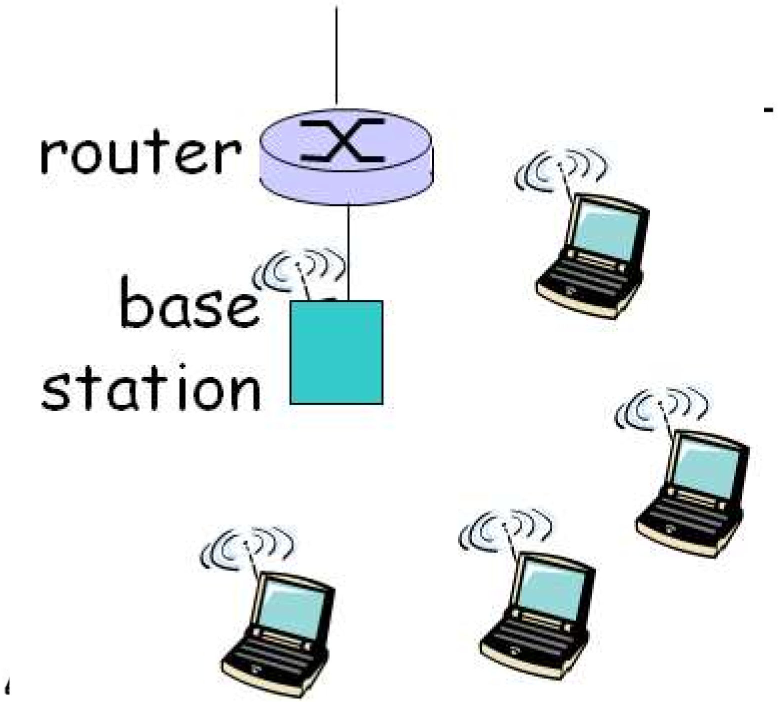
- System interconnection
  - Bluetooth (蓝牙)
  - In the simplest form, system interconnection networks use the master-slave paradigm
- Wireless LANs
  - Wifi: IEEE802.11
- Wireless MANs
  - WiMax： IEEE802.16
- Wireless WANs
  - GPRS, 3G, 4G, 5G
  - ……

System interconnection vs. WLAN



(a)

Bluetooth configuration



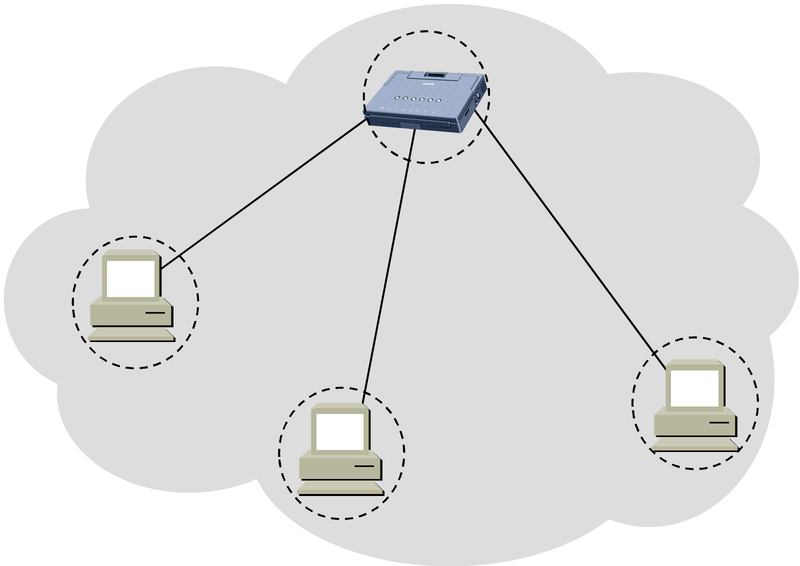
(b)

Wireless LAN

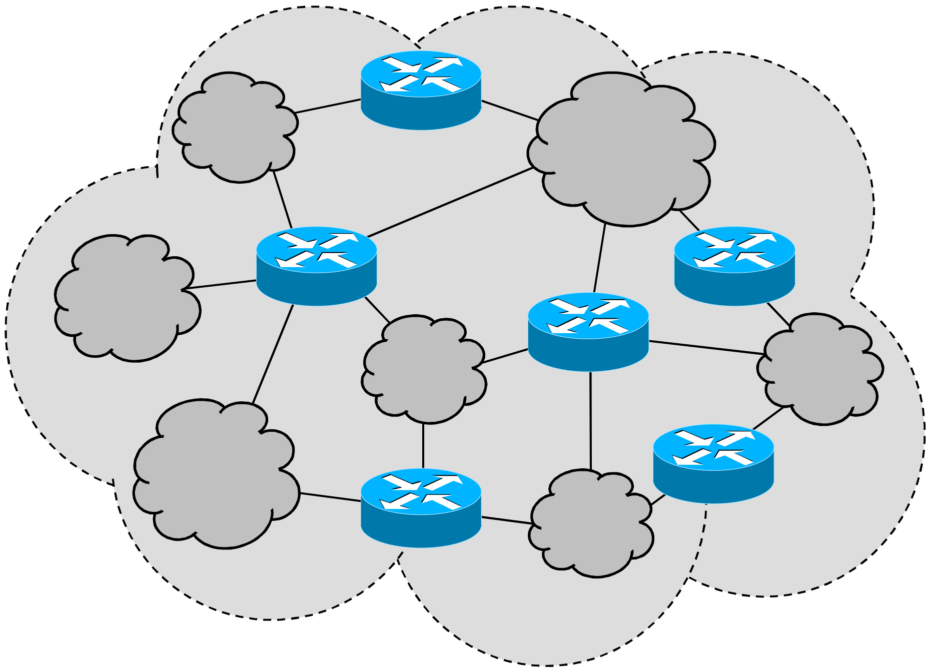
[Kurose]

Internetworks: internet（互联网）

- internet: network of networks
- Internet（因特网） is the unique world-wide internet



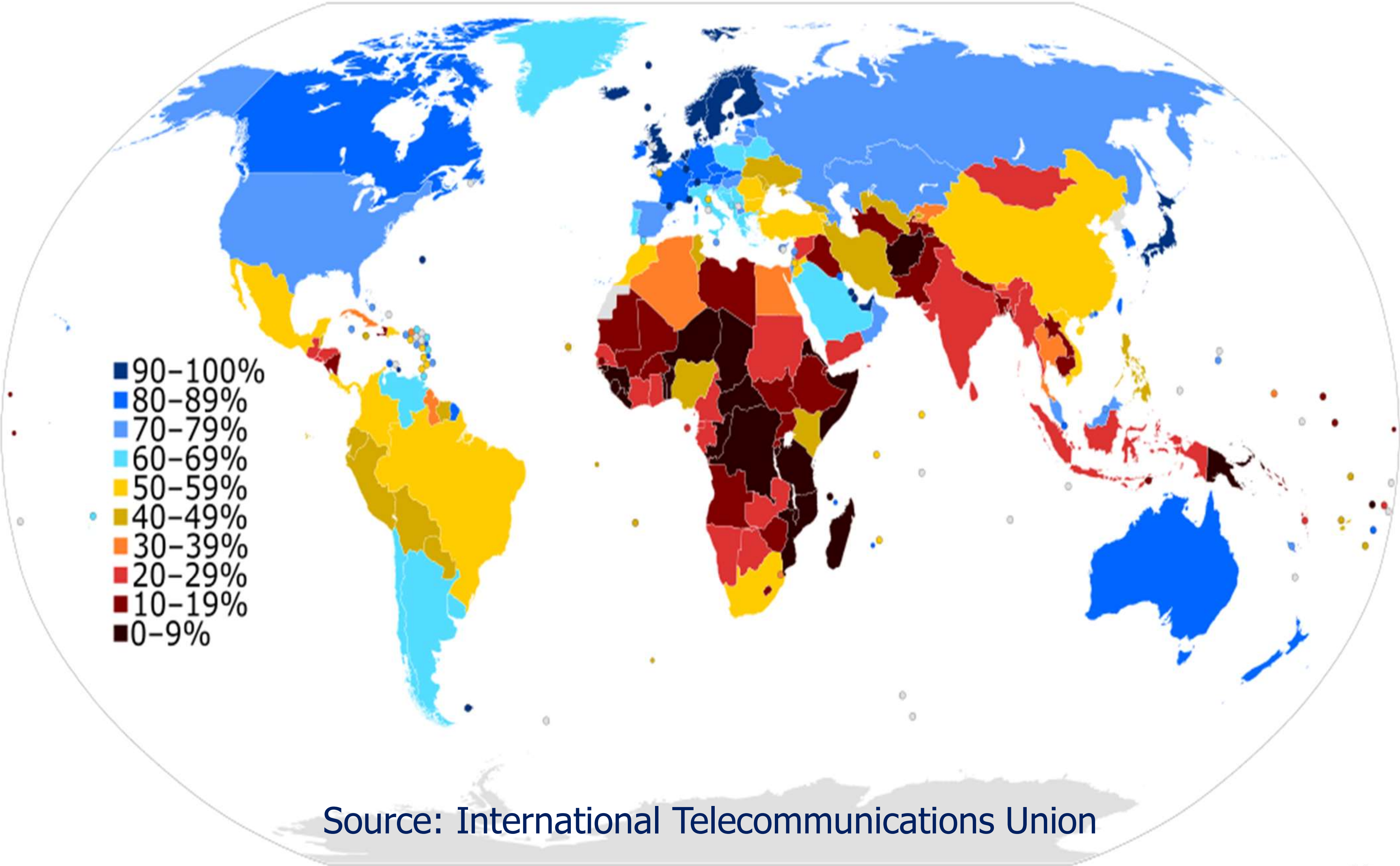
A computer network



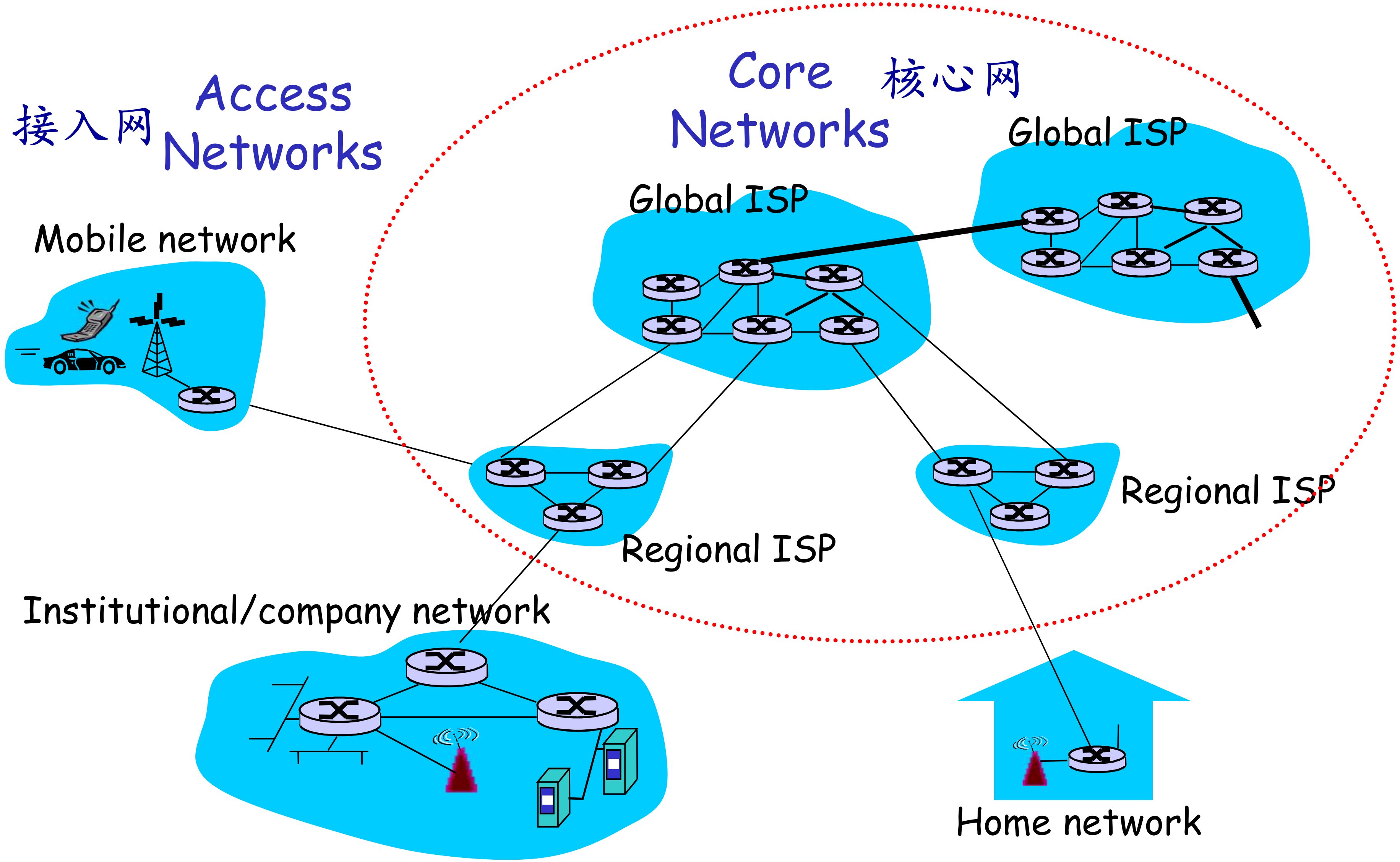
internet

[谢]

Internet: users in 2015 as a percentage of a country's population

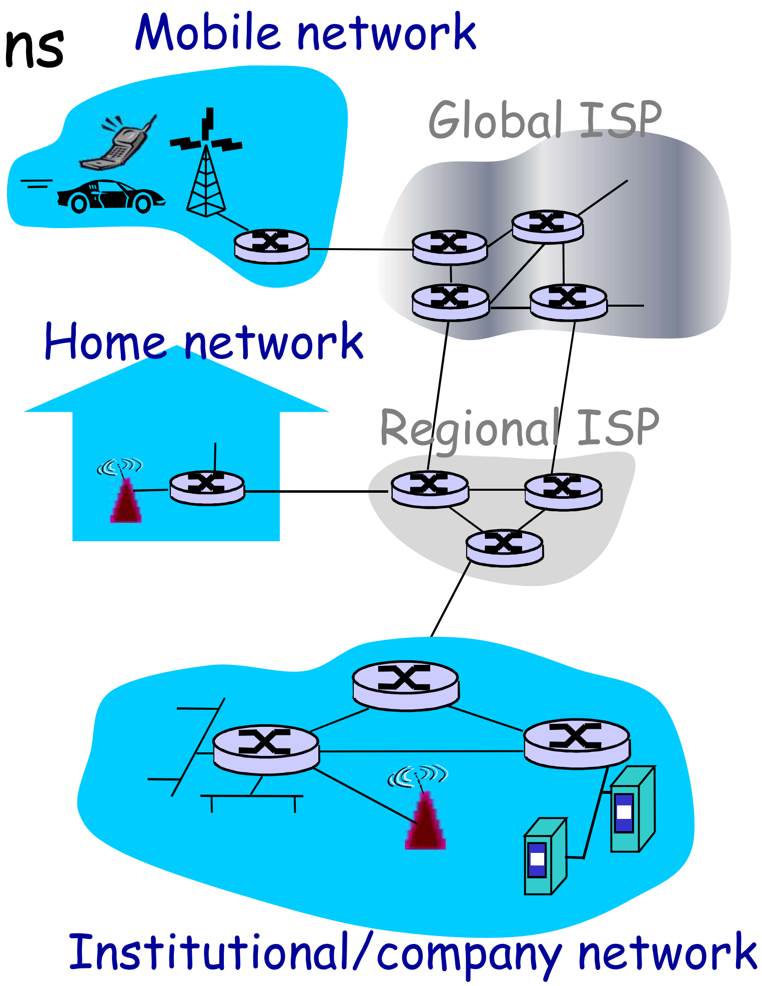


Classification on position in interconnected networks



Access Networks

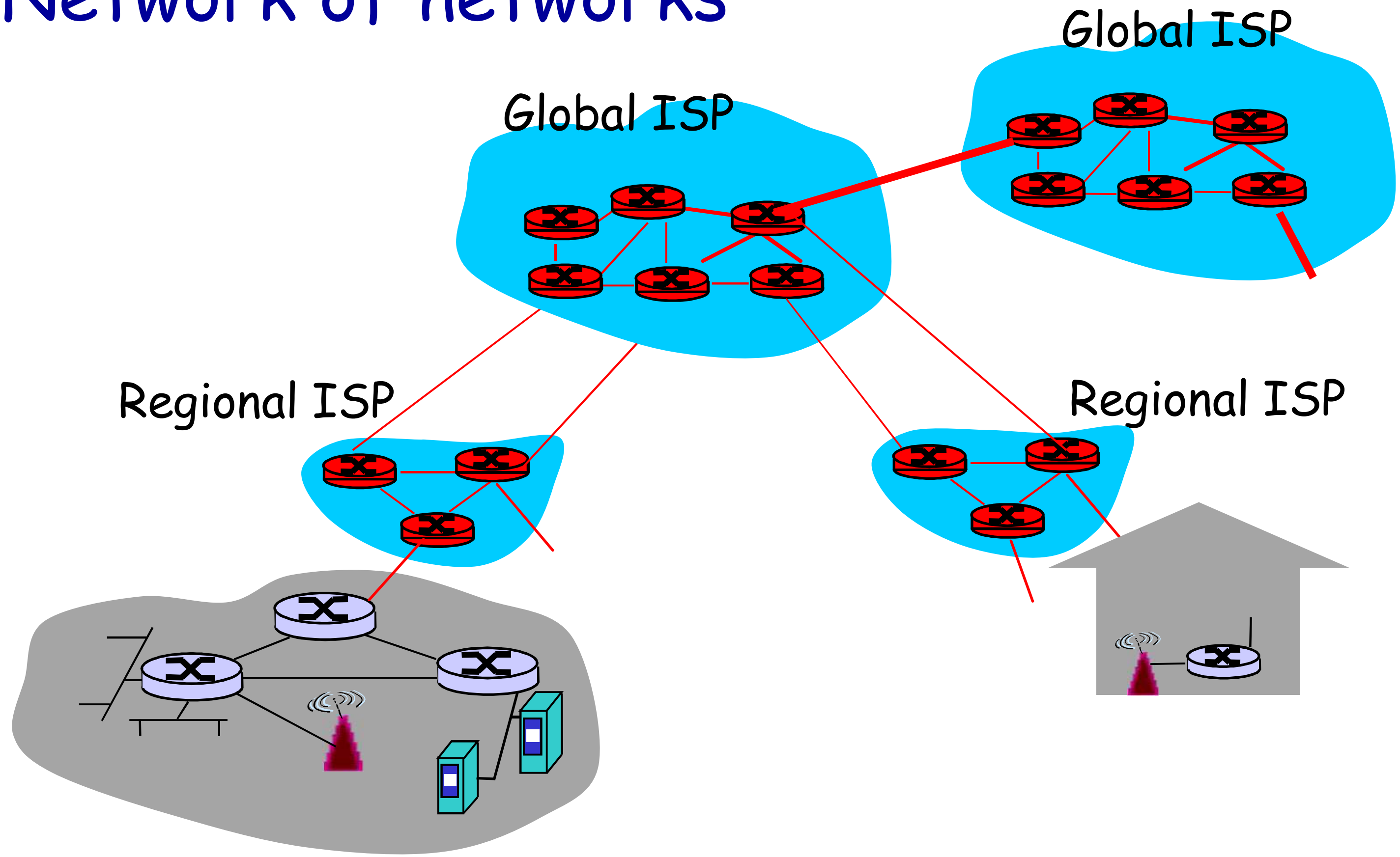
- Hosts
  - providing network applications
- Communication links
  - Mainly copper and radio
- Devices(Nodes)
  - Routers, switches, modems
- Networks types
  - LAN, WLAN
  - PSTN (ADSL, dialup)
  - FTTH
  - Cable TV (HFC)
  - GPRS, 3G, 4G, 5G
  - ……



[Kurose]

Core Networks

- Mesh of interconnected routers
- Network of networks



[Kurose]

Q: How data is transferred over the networks?

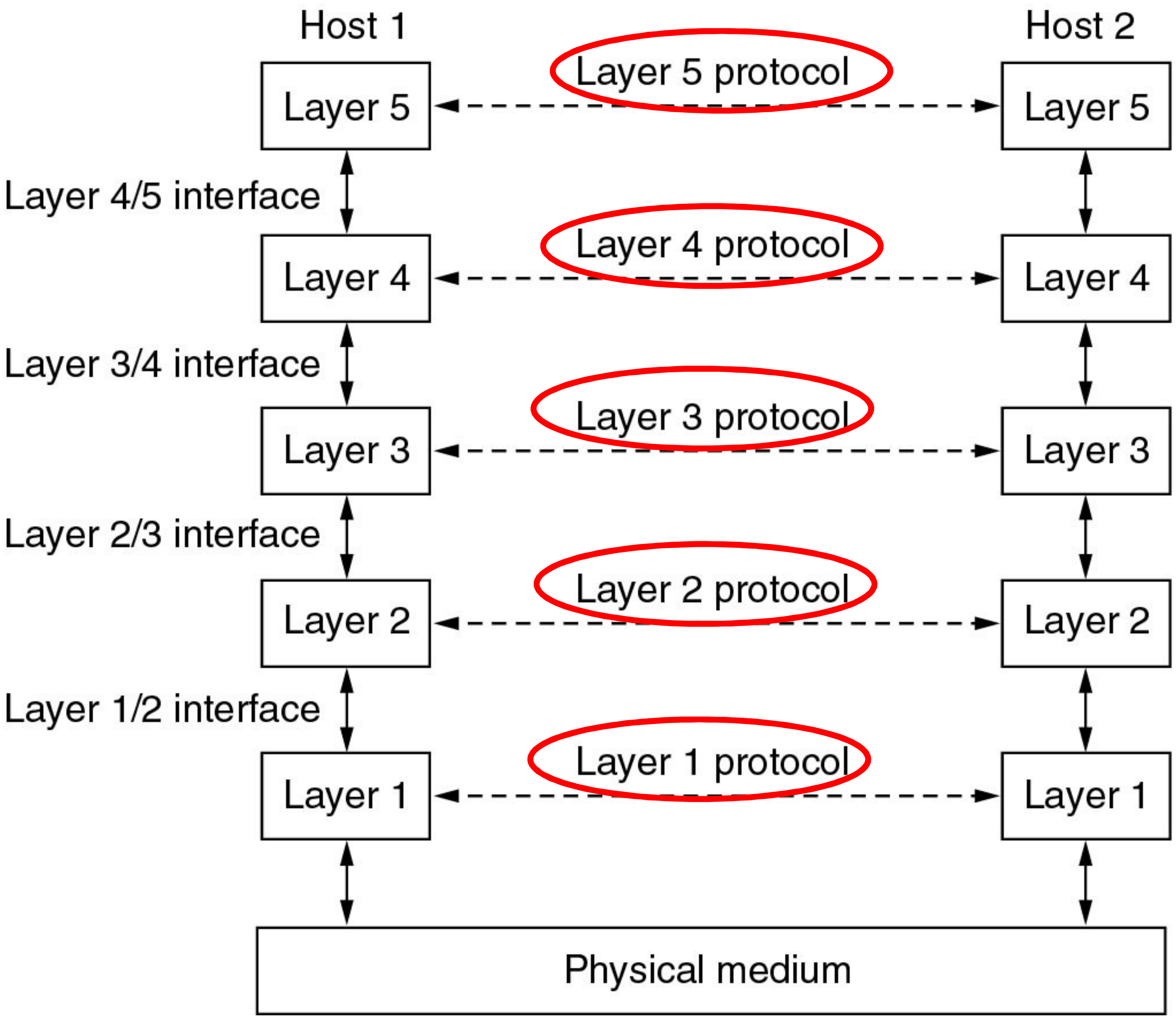
Outline

- What is a Computer Network?
- What can we do with Computer Networks?
- Categories of Computer Networks
- *Network architecture and protocols*
  - Layering Architecture
  - What is Protocol?
  - Protocol, service and interface
- Reference Models
- Example Networks
- Network Standardization

Network architecture (网络体系结构)

- Networks are organized as a stack of layers
  - Reduce design complexity
  - The purpose of each layer is to offer certain services to the higher layers, shielding those layers from the details of how the offered services are actually implemented
  - Each layer is a kind of virtual machine, offering certain services to the layer above it
- Object-oriented programming
  - A particular piece of software (or hardware) provides a service to its users, but keeps the details of its internal state and algorithms hidden from them

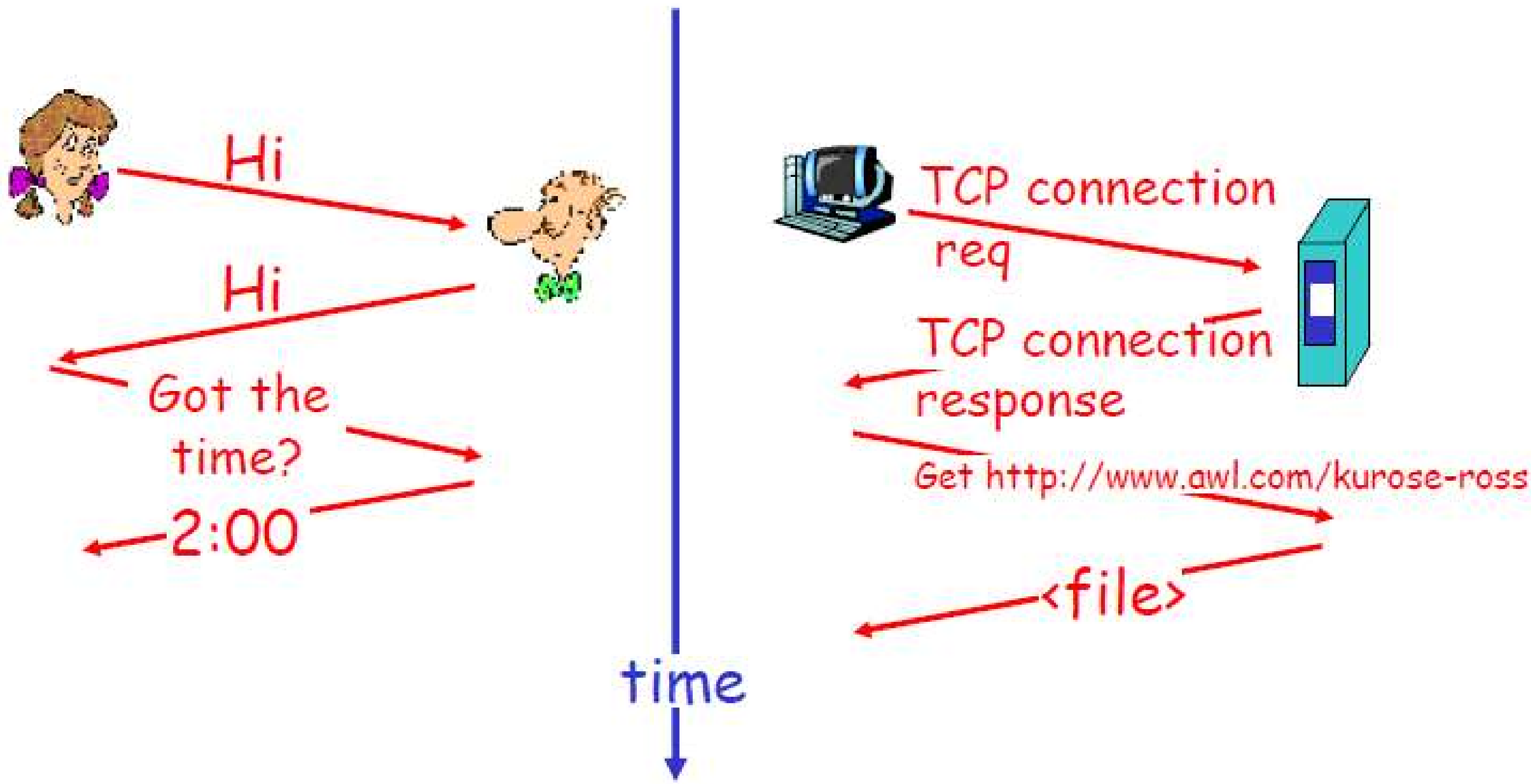
Layers, Peers, Protocols, and Interfaces



Layers, protocols, and interfaces

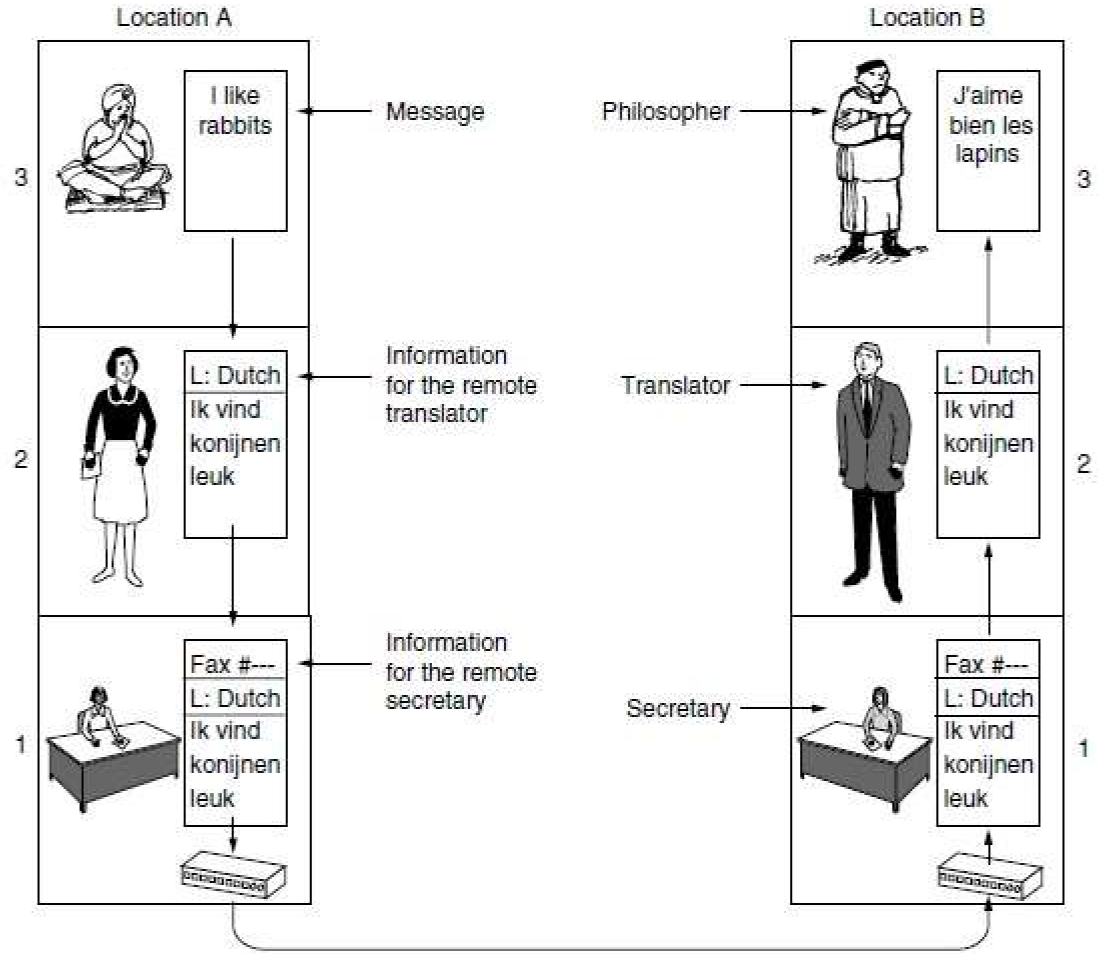
- Protocol(协议)
  - An agreement between the communicating parties on how communication is to proceed
- Peers （对等实体）
  - The entities comprising the corresponding layers on different machines
  - The peers may be processes, hardware devices
- Interface （接口）
  - Defines which primitive operations and services the lower layer makes available to the upper one
- Network architecture（网络体系结构）
  - A set of layers and protocols
  - Neither the details of the implementation nor the specification of the interfaces is part of the architecture
- Protocol Stack（协议栈）
  - A list of protocols used by a certain system

a Human Protocol vs. a Network Protocol



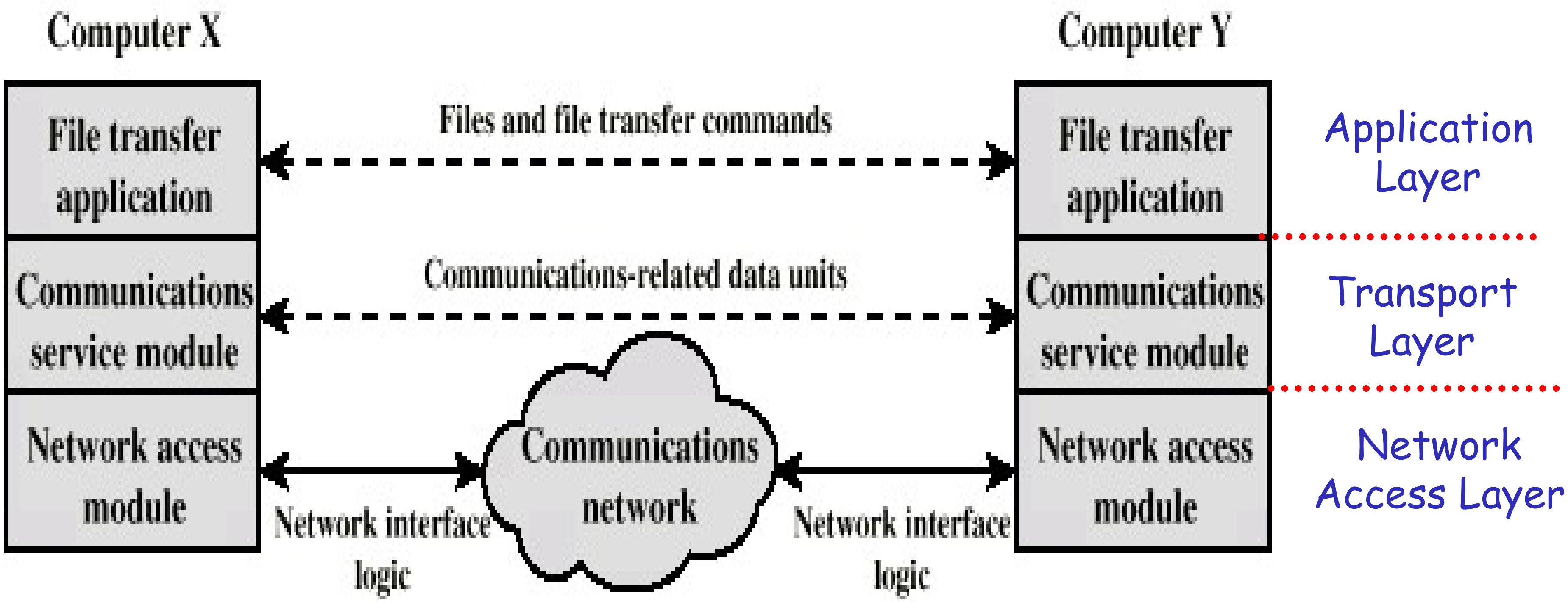
[Kurose]

3 layers: Philosopher-Translator-Secretary



- Interfaces
- Protocols
  - Translator: A neutral language(Dutch) known to both of them
  - Secretary: communication method

Simplified Network Application Example: 3 Layers



[Stallings]

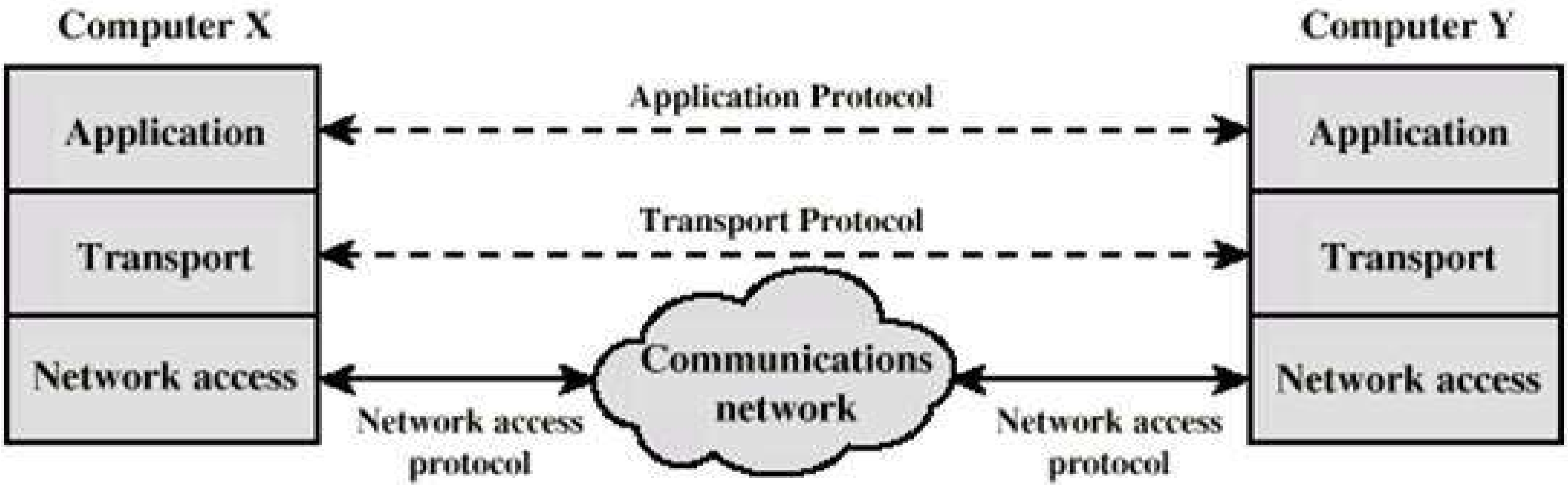
3 layers example: functions in each layer (1)

- Network access layer
  - Exchange of data between computer and network
  - Sending computer provides destination address
  - Dependent on type of network used (LAN, ADSL etc.)
- Transport layer
  - Reliable data exchange
  - Independent of network being used
  - Independent of application

3 layers example: functions in each layer (2)

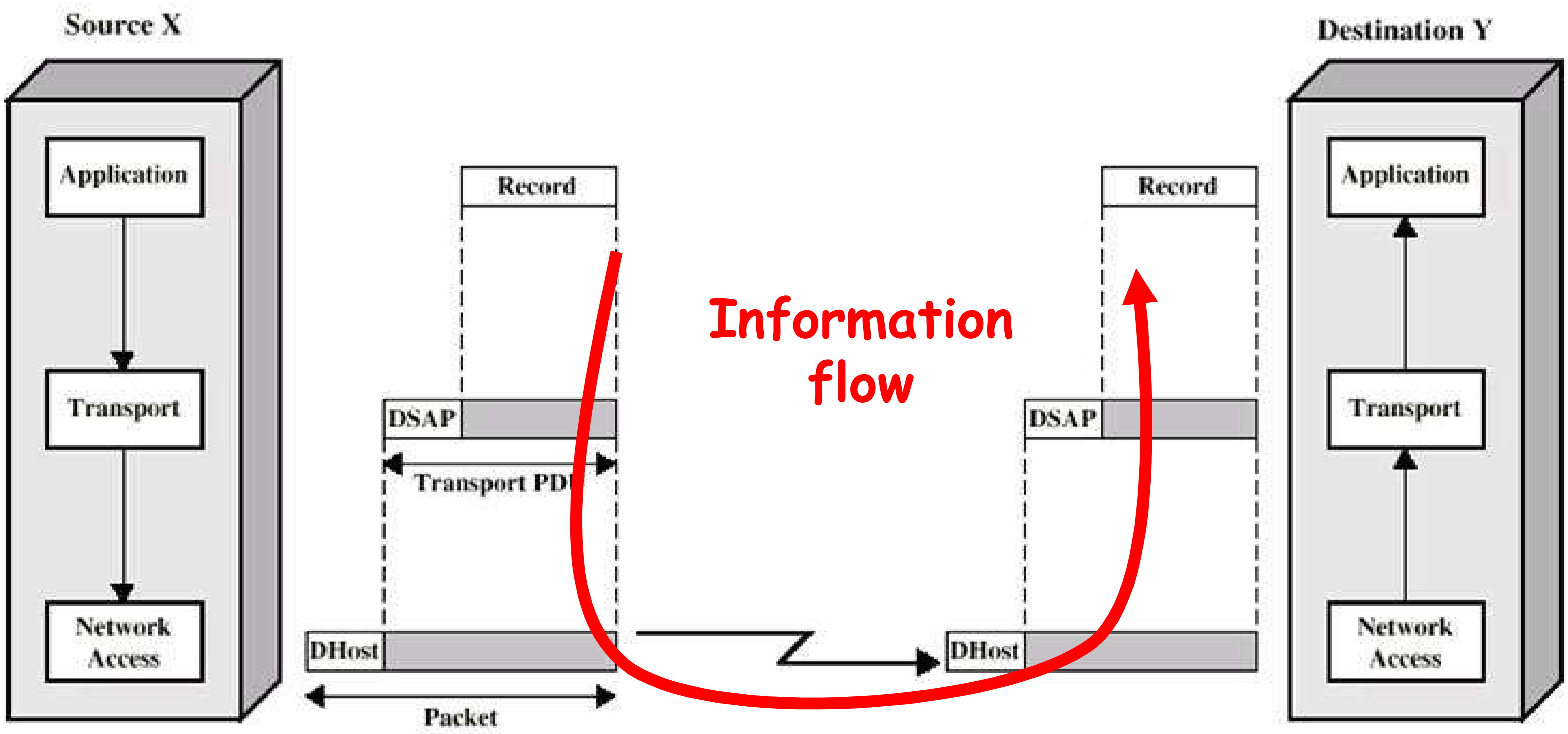
- **Application layer**
  - Support for different user applications
  - e.g. e-mail, file transfer, web browsing…

3 layers architecture: protocols



[Stallings]

3 layers example: operation of the architecture



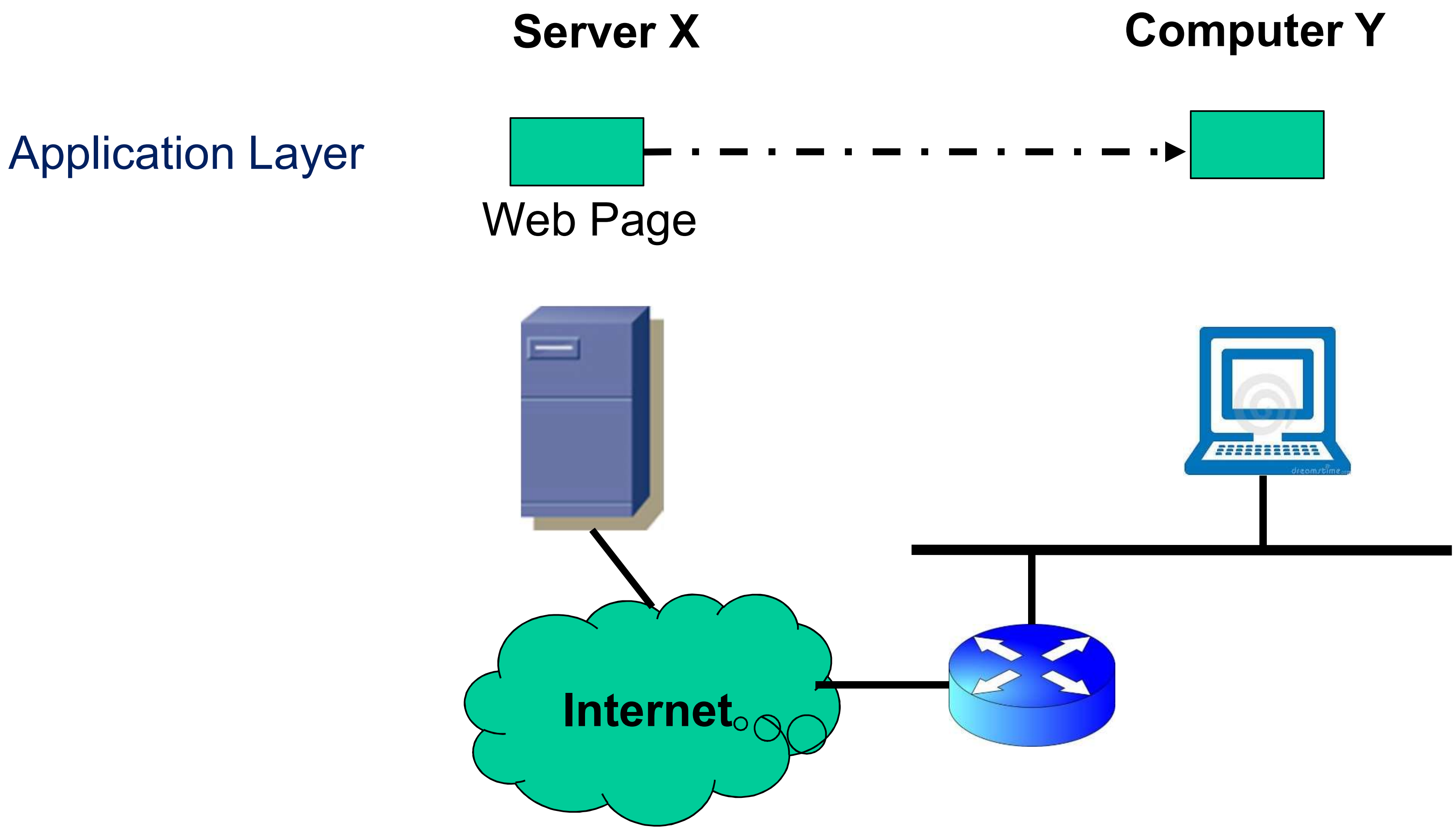
Encapsulation(封装)

DSAP: Destination Service Access Point, Address in transport layer

DHost: Network address of destination

[Stallings]

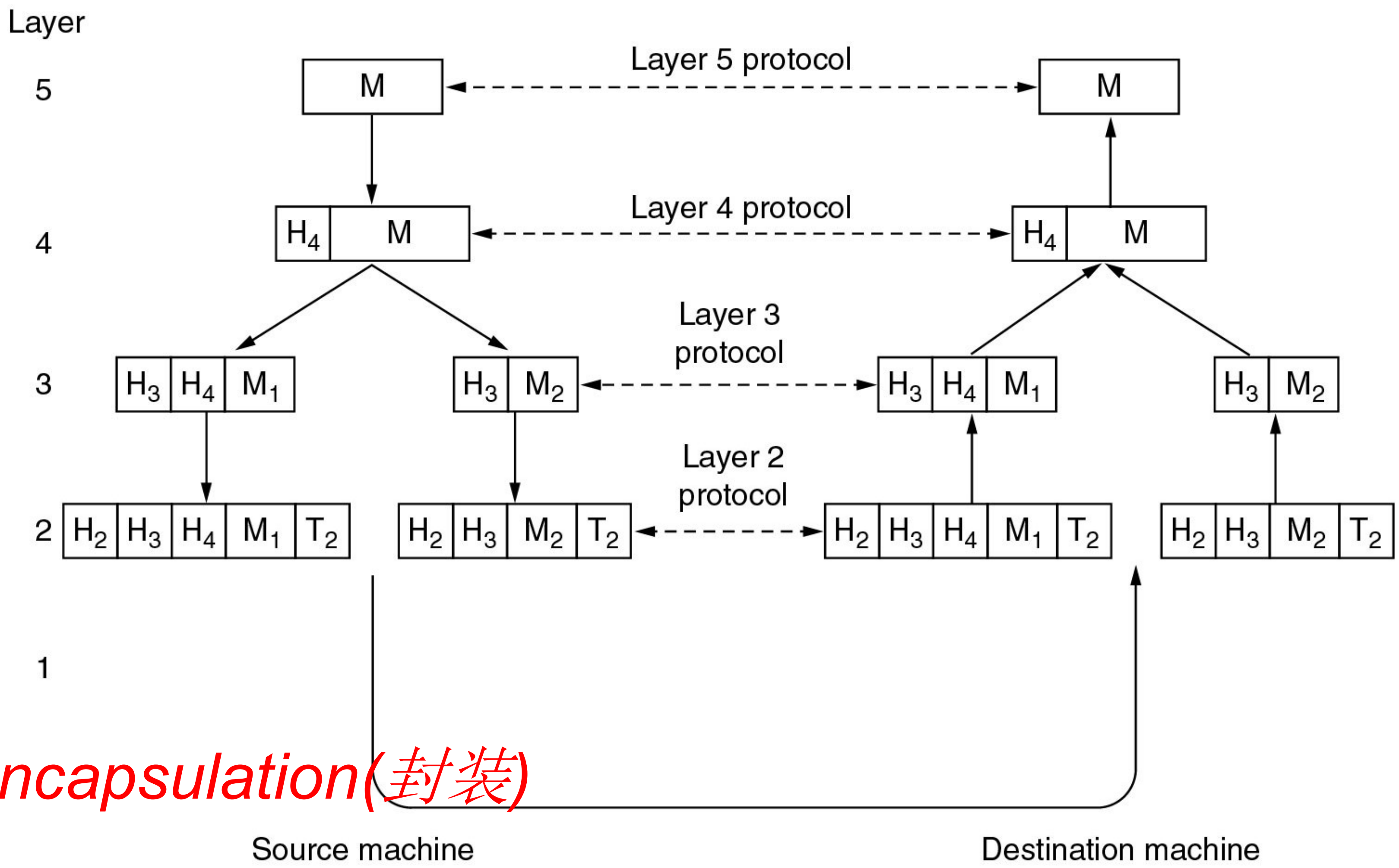
How many layers are good and enough?



5-layer may be OK

Information flow

Example information flow supporting virtual communication in layer 5.



Encapsulation(封装)

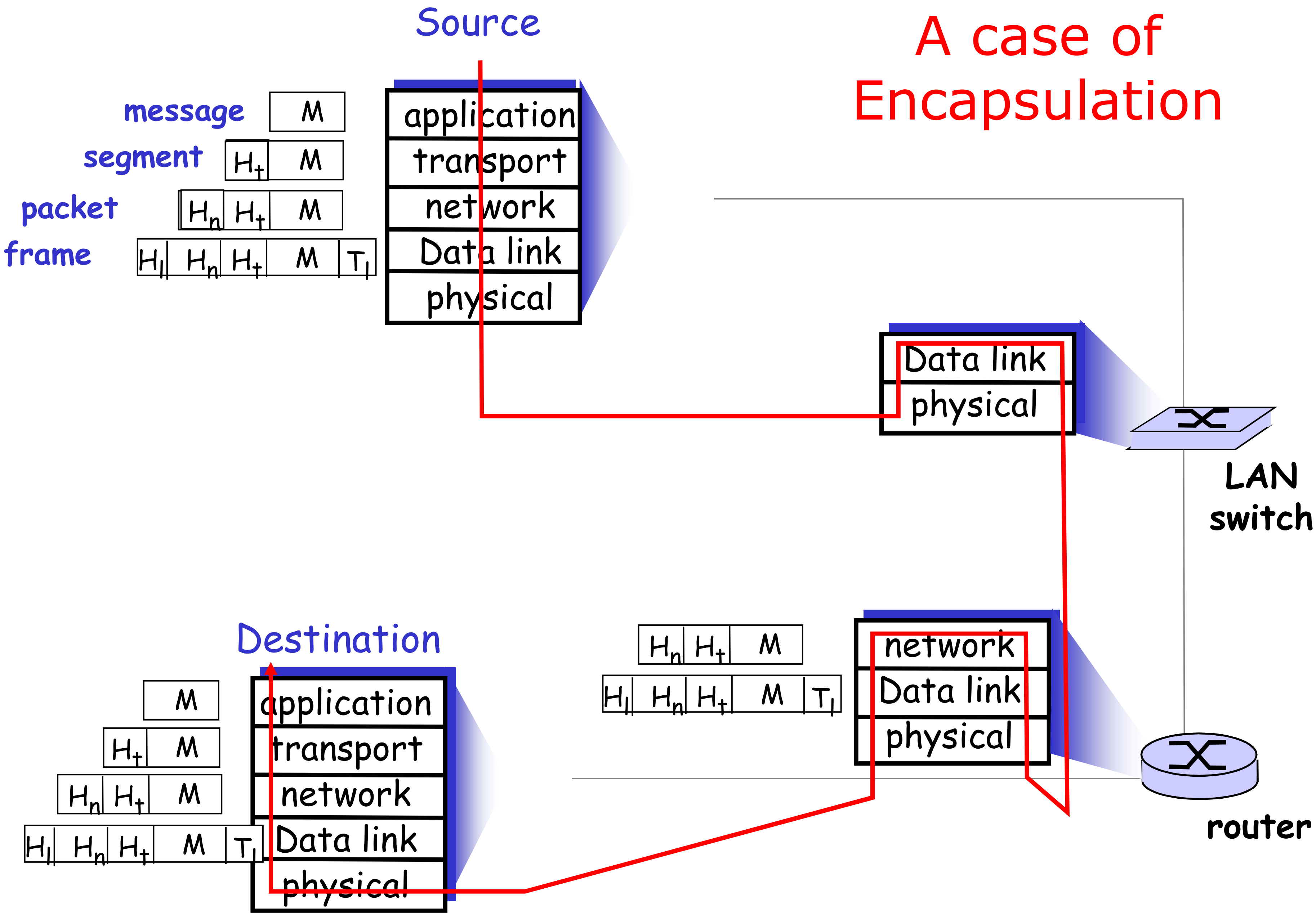
Virtual Communication of Layer 5 Peers(1)

- Layer 5
  - A message, M, is produced by an application process and given to layer 4 for transmission
- Layer 4
  - Puts a header in front of the message to identify the message and passes the result to layer 3
  - Header: control information, such as sequence numbers (In some layers, headers can also contain sizes, times, and other control fields)

Virtual Communication of Layer 5 Peers(2)

- Layer 3
  - A limit to the size of messages transmitted in the layer 3 protocol
  - Break up the incoming messages into smaller units (packets)
  - Put a layer 3 header to each packet
  - Decides which of the outgoing lines to use and passes the packets to layer 2
- Layer 2
  - Adds a header & trailer to each piece
  - Gives the resulting unit to layer 1
- Layer 1
  - Physical transmission

A case of Encapsulation



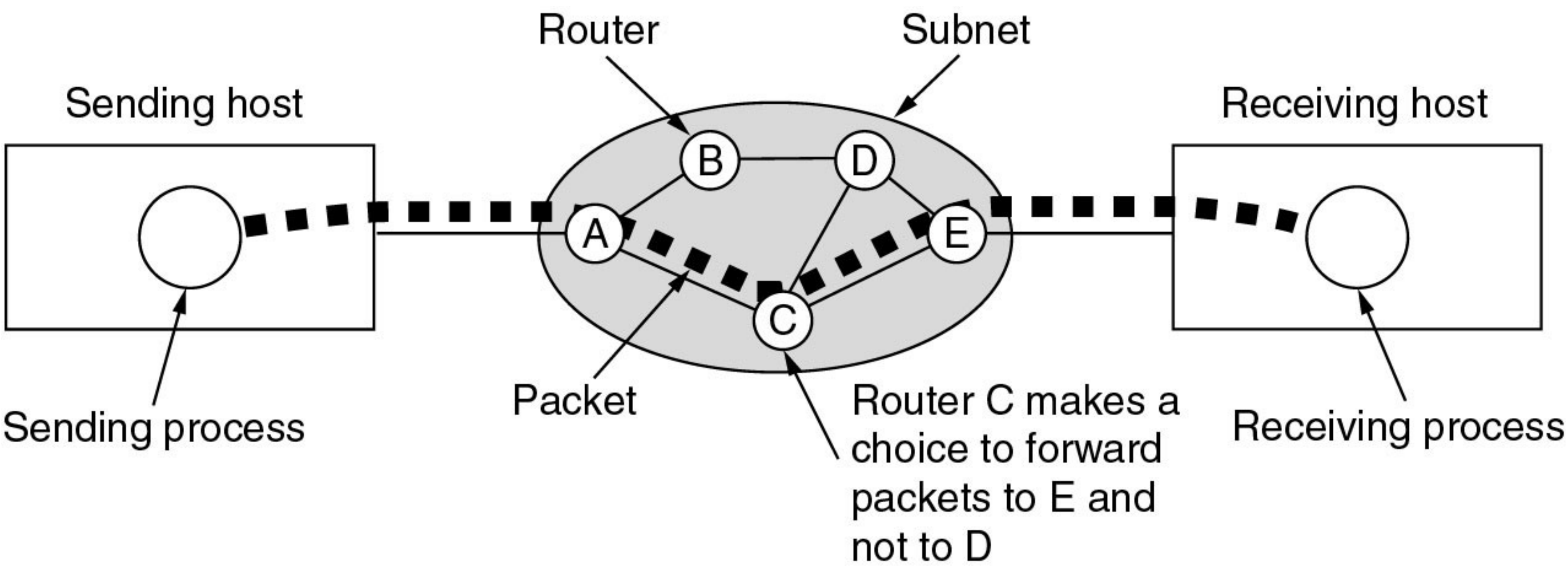
[Kurose]

Design Issues for the Layers

- Reliability
  - Error detection and error correction （差错检测和恢复）
  - Routing: to find a path through a netwrok （路由选择）
- Network evolution
  - Protocol layering: dividing the overall problem and hiding implementation details
  - Addressing or naming: identifying senders and receivers
  - Internetworking（网络互通性）
  - Scalability(可扩展性)
- Resource allocation
  - Statistical multiplexing（统计复用：按需分配）
  - Flow Control(流量控制): A fast sender may swamp a slow receiver with data
- QoS（服务质量）
- Security

Routing

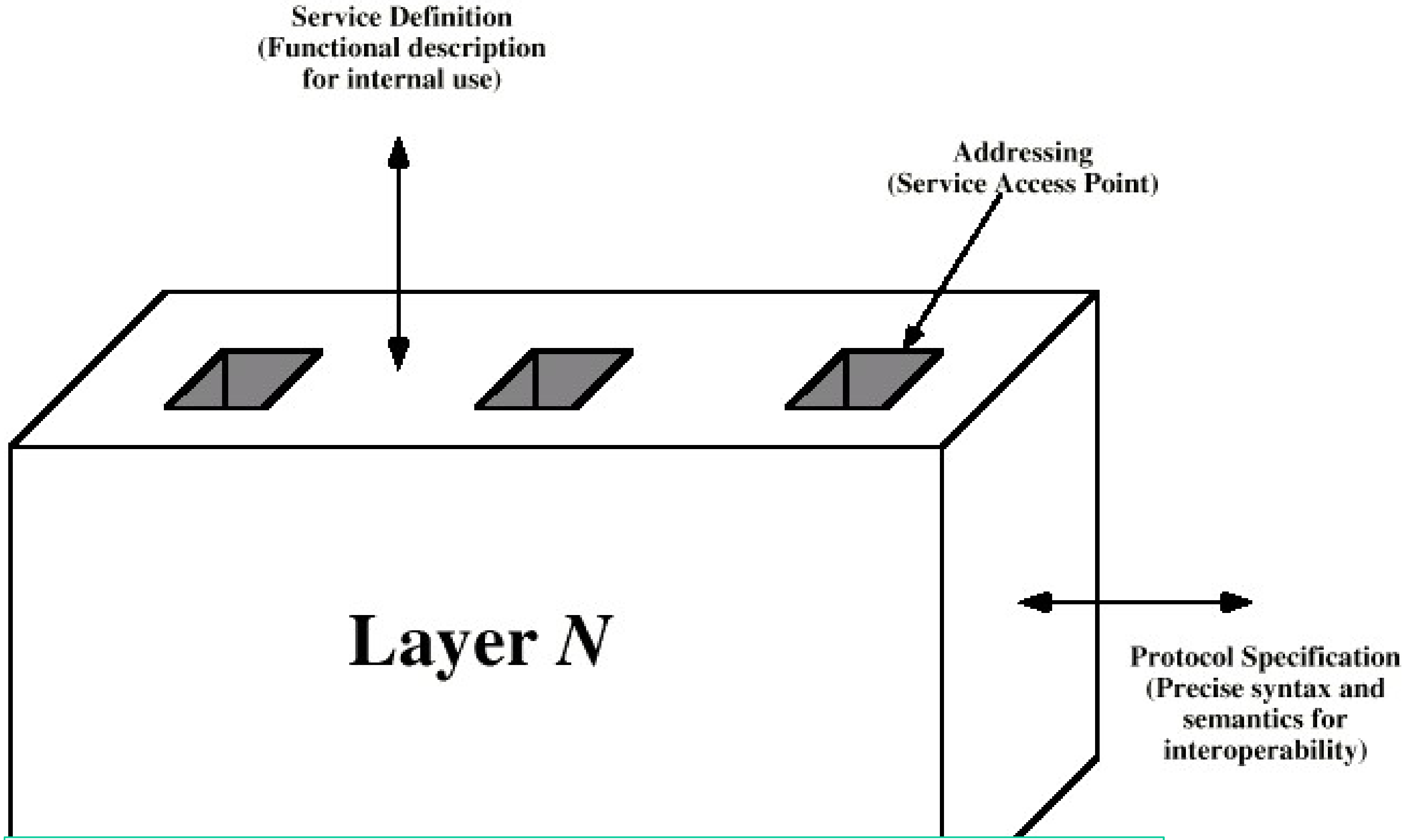
- When multiple paths exist, a route must be chosen



Review of Network Architecture

- 为什么要分层? 简化设计和实现，便于互连互通
- 网络体系结构包括：分哪几层？每层的功能是什么？有哪些协议？
- 每一层的对等实体之间进行通信，通信要遵守协议
- 只有最底层是实际通信，其它各层都是虚拟通信
- 数据流向(Information Flow)：U形，发送系统自顶向下，最底层实际传输数据，接收系统自底向上
- 封装：某层实体在上一层交付的数据前面（可能也在后面）加上自己的控制信息，构成本层的数据包
  - 这些控制信息是由协议定义的

Layer Specific Standards

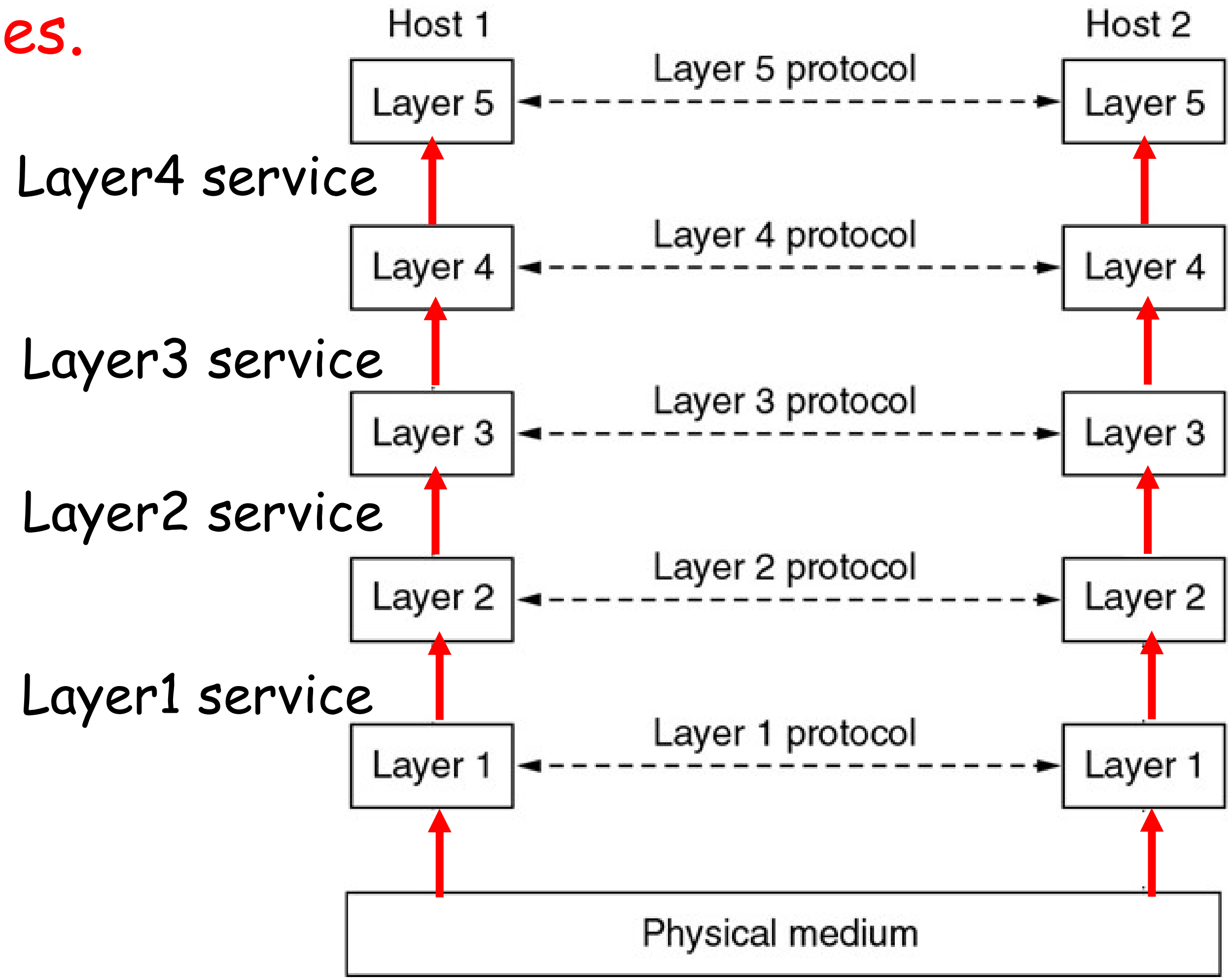


Syntax：语法，协议消息的格式
Semantics：语义，消息的字段区某个值的含义

[Stallings]

Services

- In network architecture, each layer provides a certain function(services) to its upper layer through interfaces.



Services: Connection-oriented(面向连接) vs. Connectionless(无连接)

	Connection Oriented	Connectionless
Ref.	Telephone System	Postal System
Features	connection is established, and resources are allocated; Then data transmission starts on connected path Finally connection is released, and resources are released	Dynamic resource allocation (allocated only when data arrives) Receiver is no need to be active on data transmission
QoS	Reliable delivery	Data may be lost, duplicated or out of sequence
Dest. address	Full address required only when setting up connection; Later, only connection ID required	Full address required for each packet/message
Applications	Large amount of data to transfer to a destination in a short time	Small amount of data

Six different types of services

	Service	Example
Connection-oriented	Reliable message stream	Sequence of pages
	Reliable byte stream	Remote login
	Unreliable connection	Digitized voice
Connection-less	Unreliable datagram	Electronic junk mail
	Acknowledged datagram	Registered mail
	Request-reply	Database query

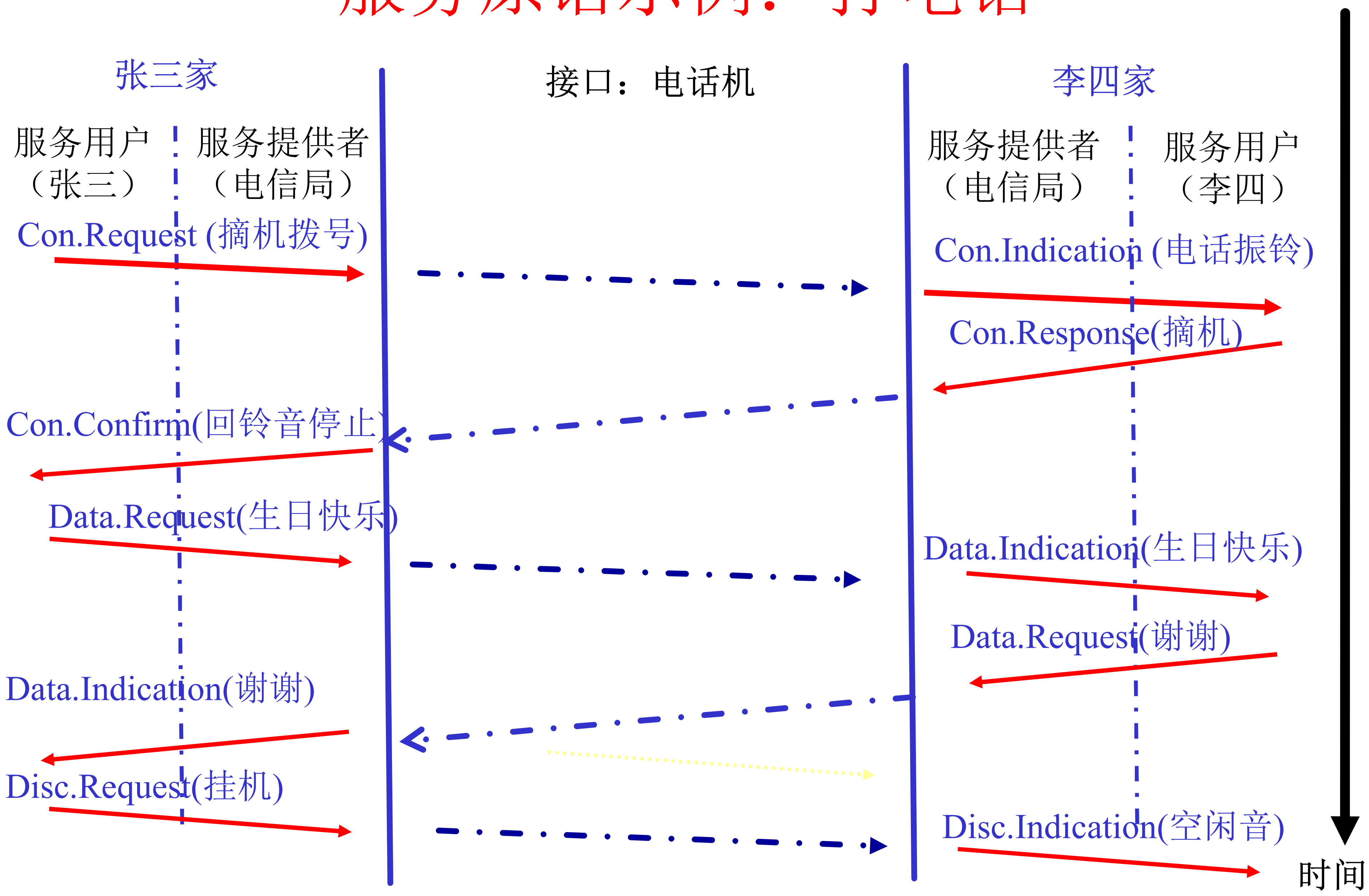
Interfaces and services

- Entity (实体)
  - Active elements in a layer, software entity or hardware entity
- Peers (对等实体)
  - Entities in the same layer on different machines
- **PDU: Protocol Data Unit (协议数据单元)**
  - Information exchanged between two peers
- Service provider and service user
  - The entities in layer n implement a service used by layer n+1, layer n is called the services provider, layer n+1 is called service user
- SAP: Service Access Point (服务访问点)
  - Layer n SAPs are the places where layer n+1 can access the services offered

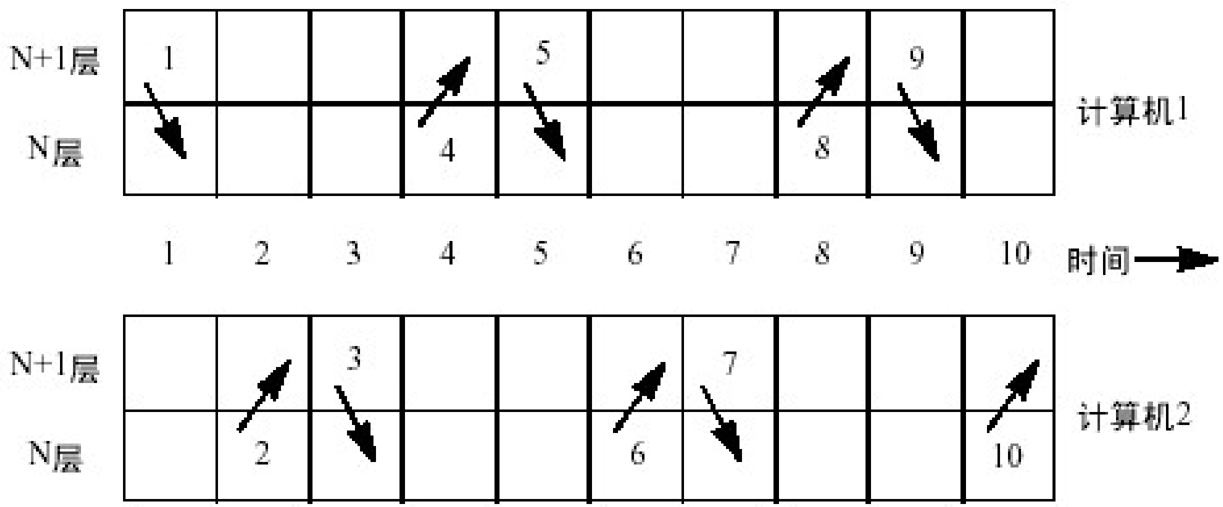
Service Primitives(服务原语): Message Style

- *A service is formally specified by a set of primitives (operations) available to a user process to access the service*
- Example
  - CONNECT.request - request a connection to be established
  - CONNECT.indication - signal the called party
  - CONNECT.response - used by the called party to accept/reject calls
  - CONNECT.confirm – tell the caller whether the call was accepted
  - DATA.request – request that data be sent
  - DATA.indication – signal the arrival of data
  - DISCONNECT.request – request that a connection be released
  - DISCONNECT.indication – signal the peer about the request

服务原语示例：打电话


张三家
接口：电话机
李四家
服务用户（张三）
服务提供者（电信局）
服务提供者（电信局）
服务用户（李四）
Con.Request (摘机拨号)
Con.Indication (电话振铃)
Con.Response(摘机)
Con.Confirm(回铃音停止)
Data.Request(生日快乐)
Data.Indication(生日快乐)
Data.Request(谢谢)
Data.Indication(谢谢)
Disc.Request(挂机)
Disc.Indication(空闲音)
时间

Service Primitives: Example



- 1. CONNECT.request
- 2. CONNECT.indication
- 3. CONNECT.response
- 4. CONNECT.confirm
- 5. DATA.request
- 6. DATA.indication
- 7. DATA.request
- 8. DATA.indication
- 9. DISCONNECT.request
- 10.DISCONNECT.indication

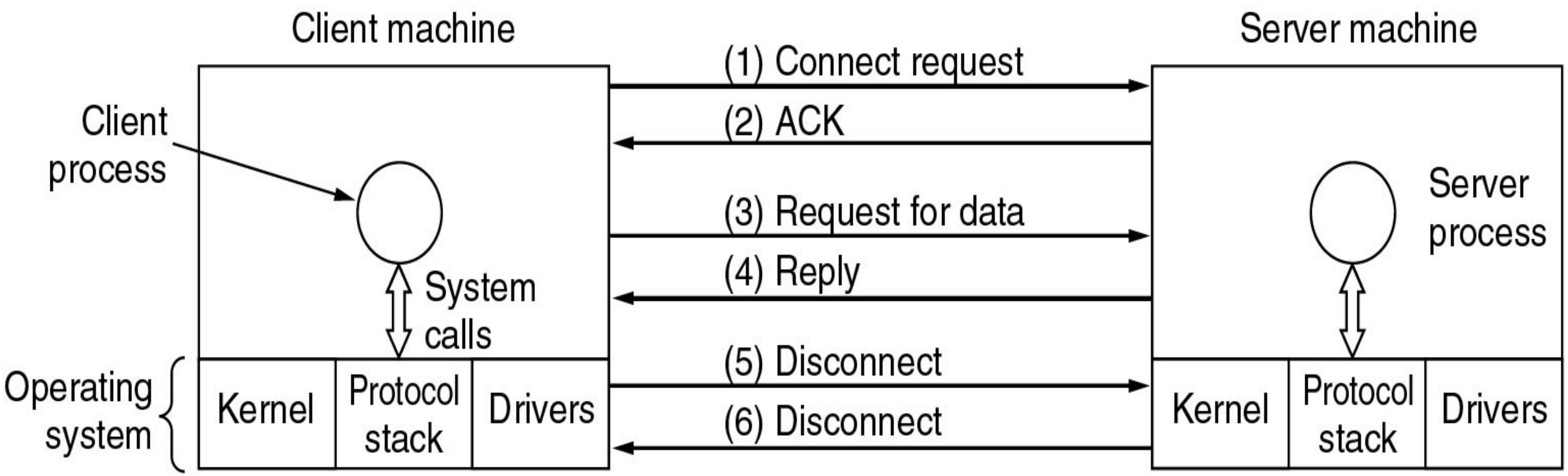
A Simple Connection-oriented Service

- If the protocol stack is located in the operating system, the primitives are normally system calls（系统调用）
  - 系统调用：由操作系统提供给用户的应用编程接口(API)
- Five service primitives for implementing a simple connection-oriented service.

Primitive	Meaning
LISTEN	Block waiting for an incoming connection
CONNECT	Establish a connection with a waiting peer
RECEIVE	Block waiting for an incoming message
SEND	Send a message to the peer
DISCONNECT	Terminate a connection

A Simple Connection-oriented Service

- Packets sent in a simple client-server interaction on a connection-oriented network.



Process: 进程，程序的一次执行

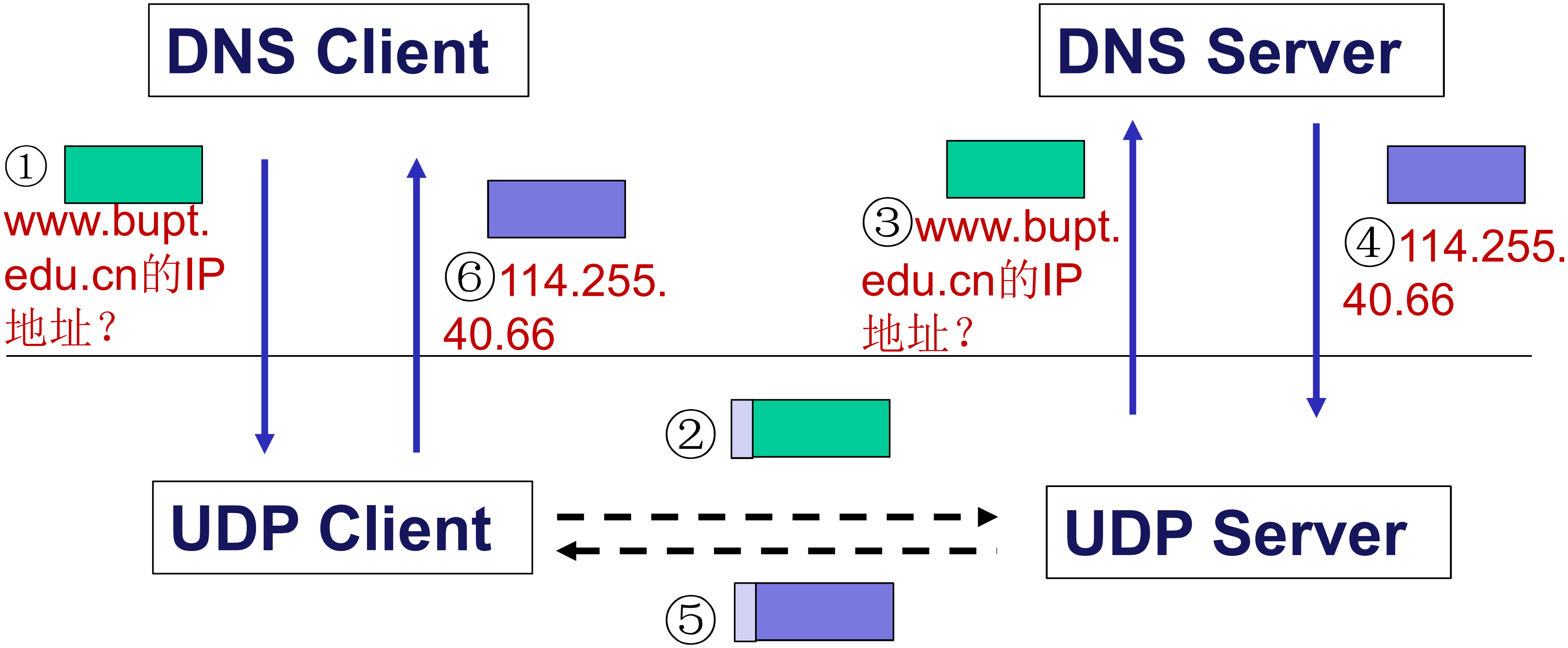
Example of protocol and service

- 场景：用户在自己的计算机上浏览北邮主页
- 分为两步：

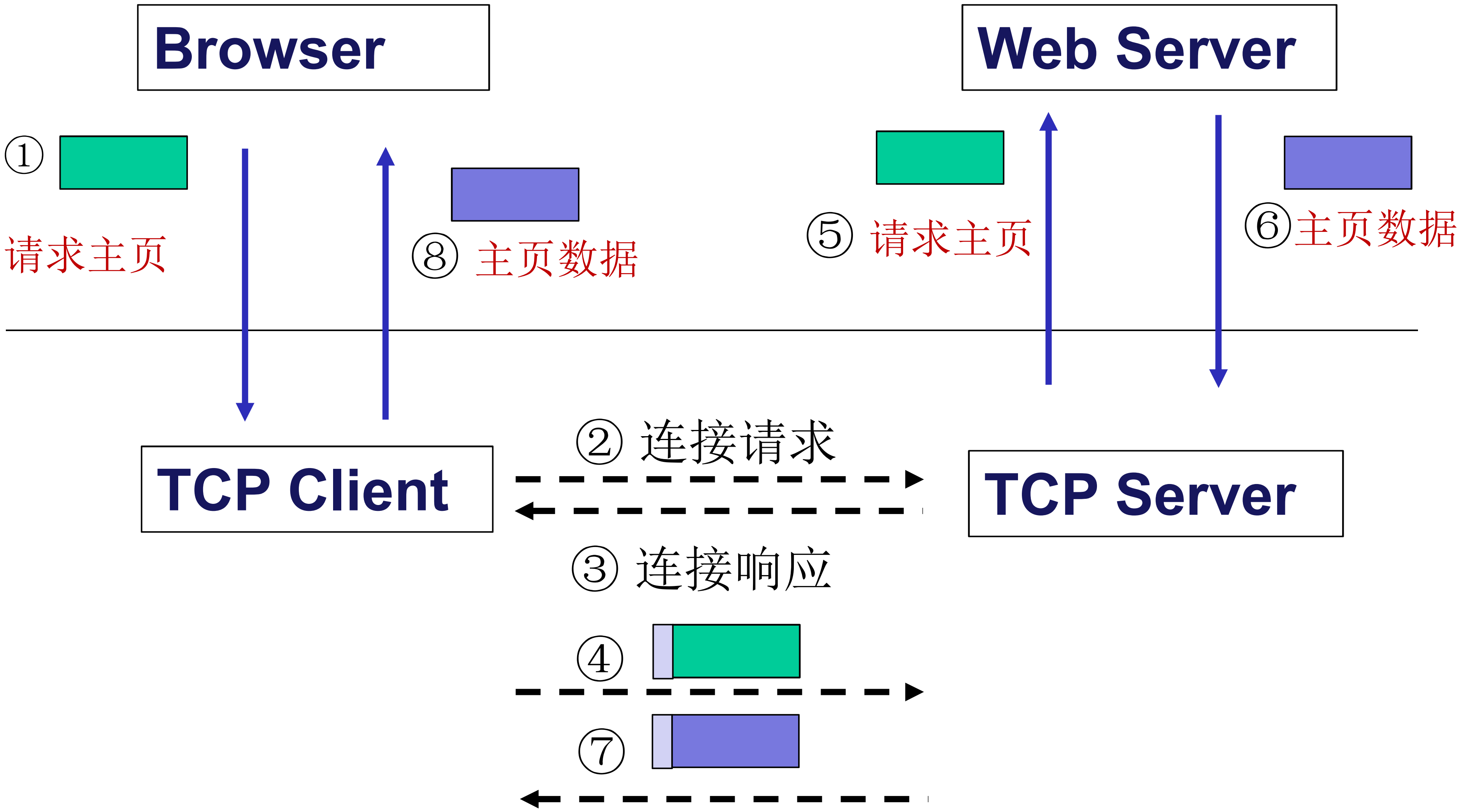
  1）用户输入域名www.bupt.edu.cn，DNS协议翻译为IP地址，下层由UDP提供无连接服务

  2）浏览器发送请求给Web服务器，服务器返回主页，下层由TCP提供面向连接的服务

Working of DNS



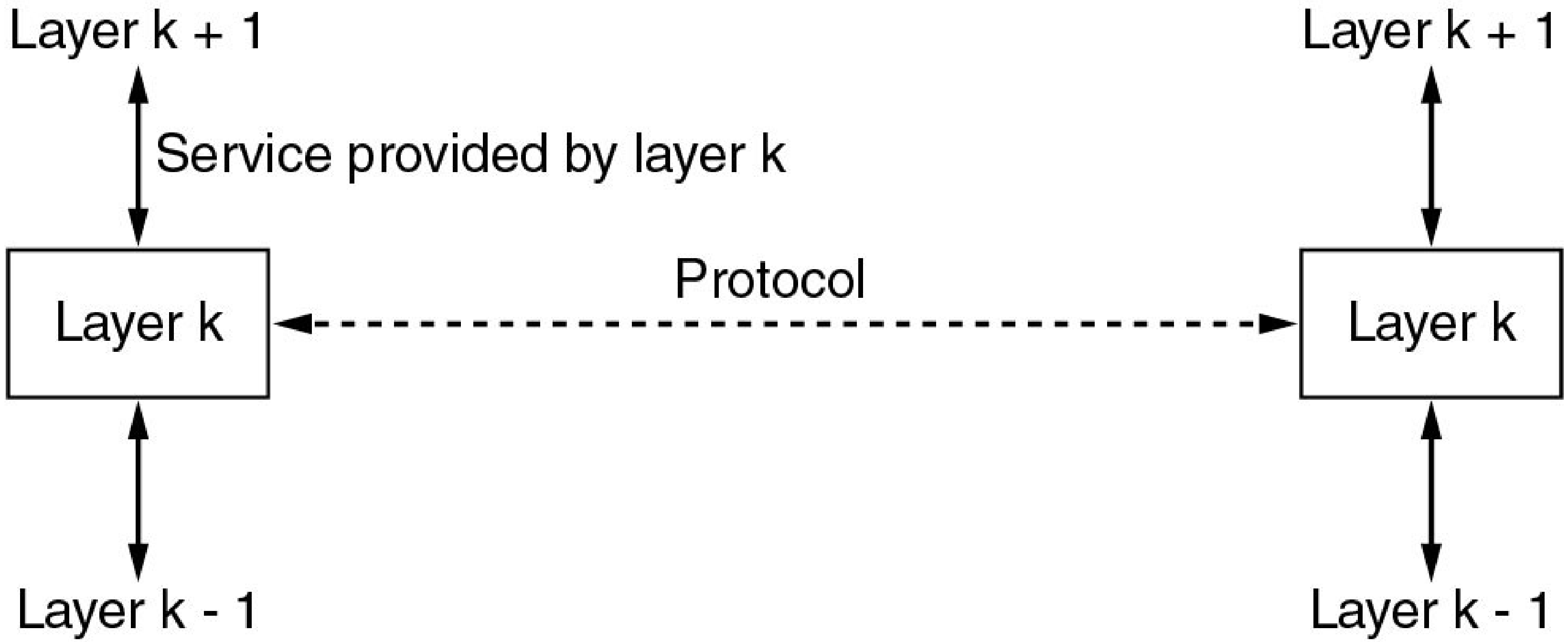
Working of Web browsing



Services vs. Protocols(1)

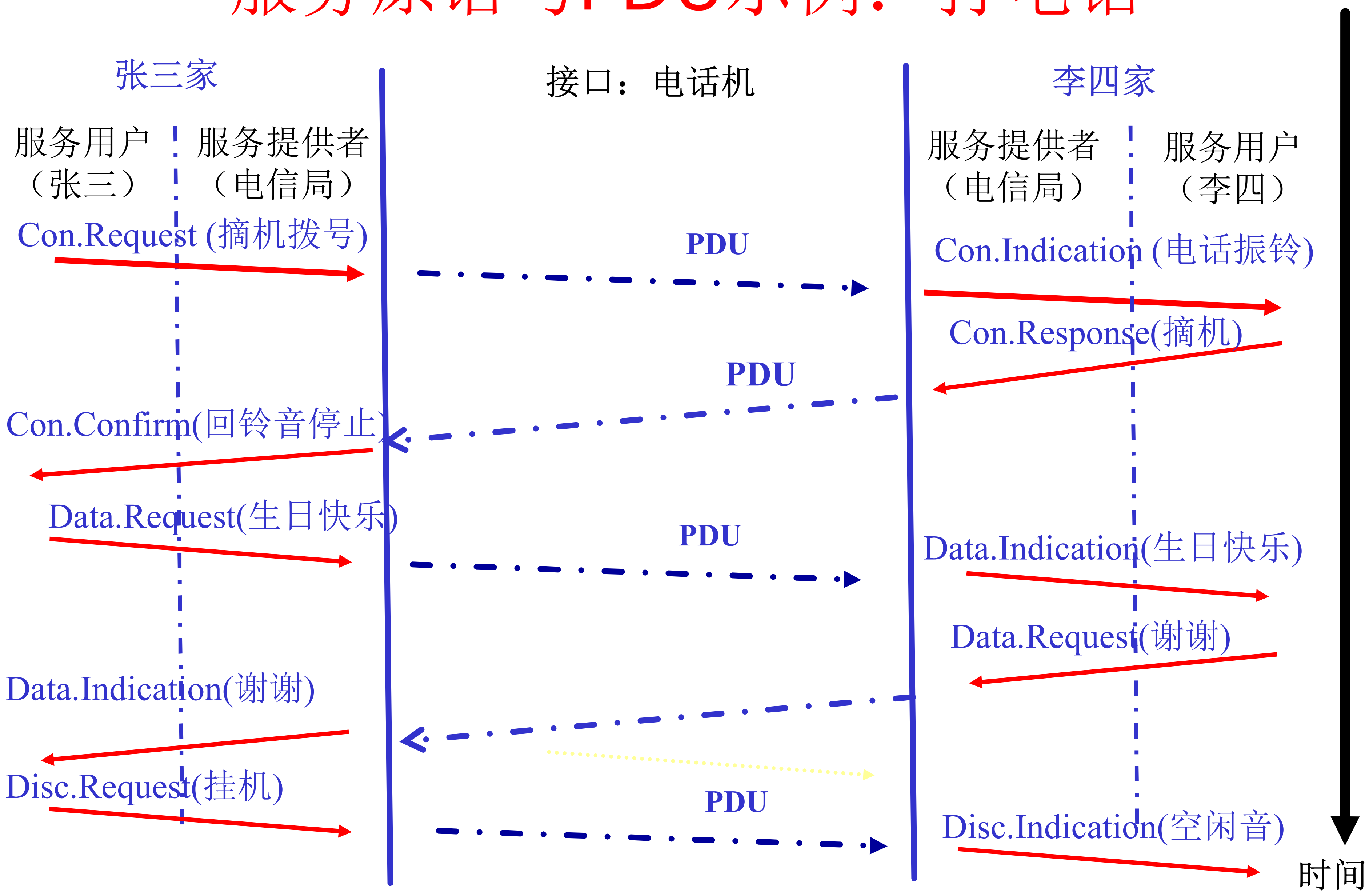
- Service
  - Defines what operations the layer is prepared to perform on behalf of its users, but it says nothing at all about how these operations are implemented
  - Relates to an interface between two layers, with the lower layer being the service provider and the upper layer being the service user
- Protocol
  - A set of rules governing the format and meaning of the messages that are exchanged by the peer entities
  - Entities use protocols to implement their service definitions
  - 服务是通过协议实现的
- Service and protocol are completely decoupled
  - They are free to change their protocols if they do not change the service

Services vs. Protocols(2)



- Entities use protocols to implement their services.
- Entities are free to change protocols at will, provided services visible to their users remain unchanged.

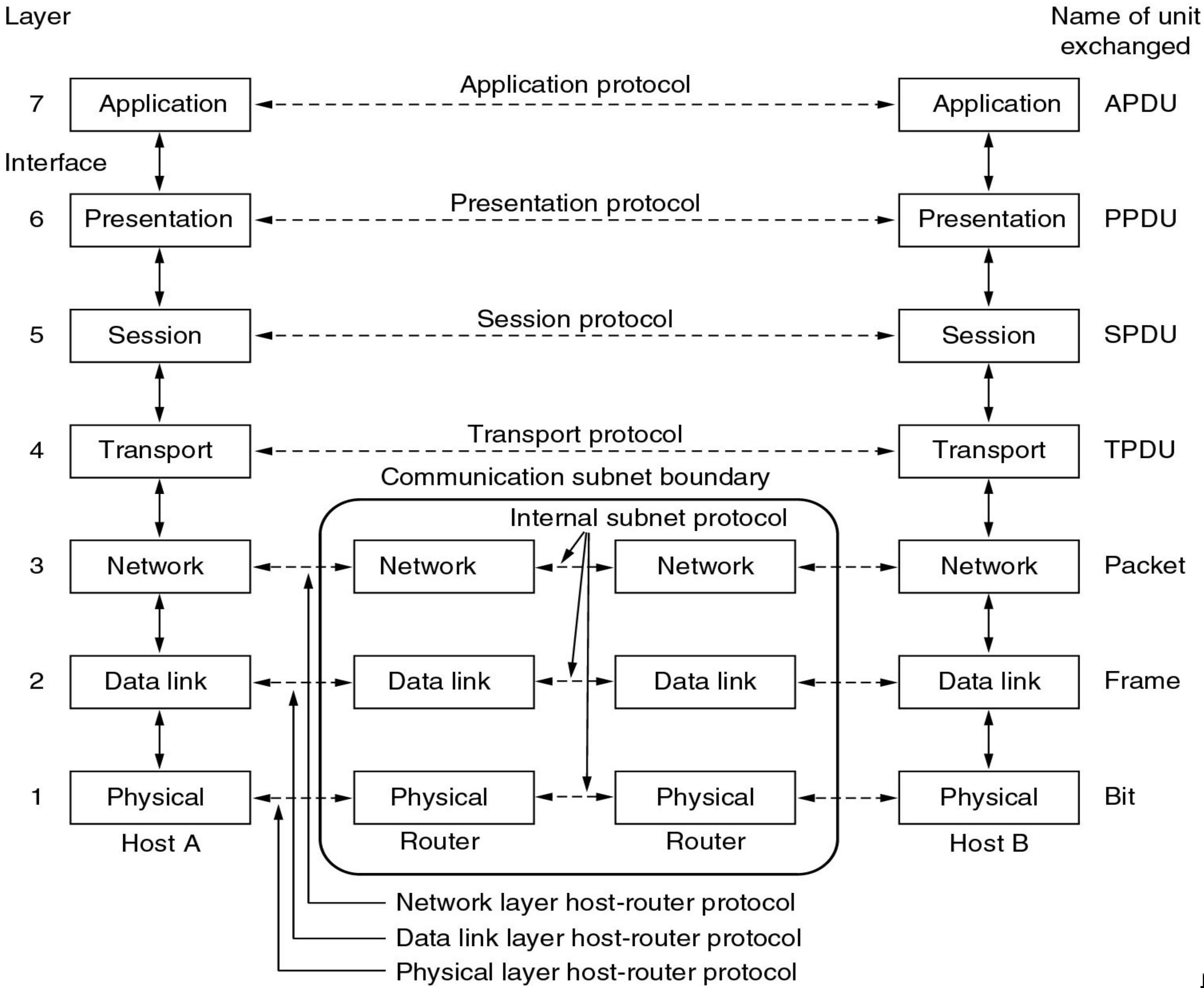
服务原语与PDU示例：打电话



Outline

- What is a Computer Network?
- What can we do with Computer Networks?
- Categories of Computer Networks
- Network architecture and protocols
- *Reference Models*
  - *OSI RM*
  - *TCP/IP Suite*
- Example Networks
- Network Standardization

The OSI reference model



Layer principles

- Each layer should perform a well-defined function
- The layer boundaries should be chosen to minimize the information flow across the interfaces

Physical layer(物理层)

- Transmitting raw bits (二进制位) over a communication channel
  - how many volts should be used to represent a 1 and how many for a 0
  - how many nanoseconds a bit lasts（数据率）
  - whether transmission may proceed simultaneously in both directions （全双工？）
  - how the initial connection is established and how it is torn down when both sides are finished

Data link layer(数据链路层)

- Transform a raw transmission facility into a logic channel
  - Sender break up the input data into frames (帧) and transmit the frames
  - If the service is reliable, the receiver confirms correct receipt of each frame by sending back an acknowledgement frame
  - Flow control(流量控制): Keep a fast transmitter from drowning a slow receiver in data
  - Broadcast networks: how to control access to the shared channel
- Point-to-Point: the protocols are between each machine and its immediate neighbors

Network layer(网络层)

- Controls operation of subnet(子网)
  - Forwarding （转发数据）
  - Routing: How packets are routed from source to destination (route table) （路由选择）
    - Highly dynamic
  - Congestion control（拥塞控制）
  - QoS (Quality of service, 服务质量)
    - delay, transit time, jitter, etc.
  - Heterogeneous networks interconnection
  - For broadcast networks(such as LAN), routing is simple

Transport Layer(传输层)

- Transport Layer: end-to-end layer
  - Accept data from above layer, split it up into smaller units if need be, pass these to the network layer, and ensure that the pieces all arrive correctly at the other end
  - Determines what type of service to provide to the session layer
    - an error-free end-to-end channel
    - transporting of isolated messages, with no guarantee about the order of delivery
    - broadcasting of messages to multiple destinations

Session Layer(会话层)

- The session layer allows users on different machines to establish sessions
- Dialog control (会话控制)
  - keeping track of whose turn it is to transmit
- Token management (令牌管理)
  - preventing two parties from attempting the same critical operation at the same time
- Synchronization (同步)
  - checkpointing long transmissions to allow them to continue from where they were after a crash

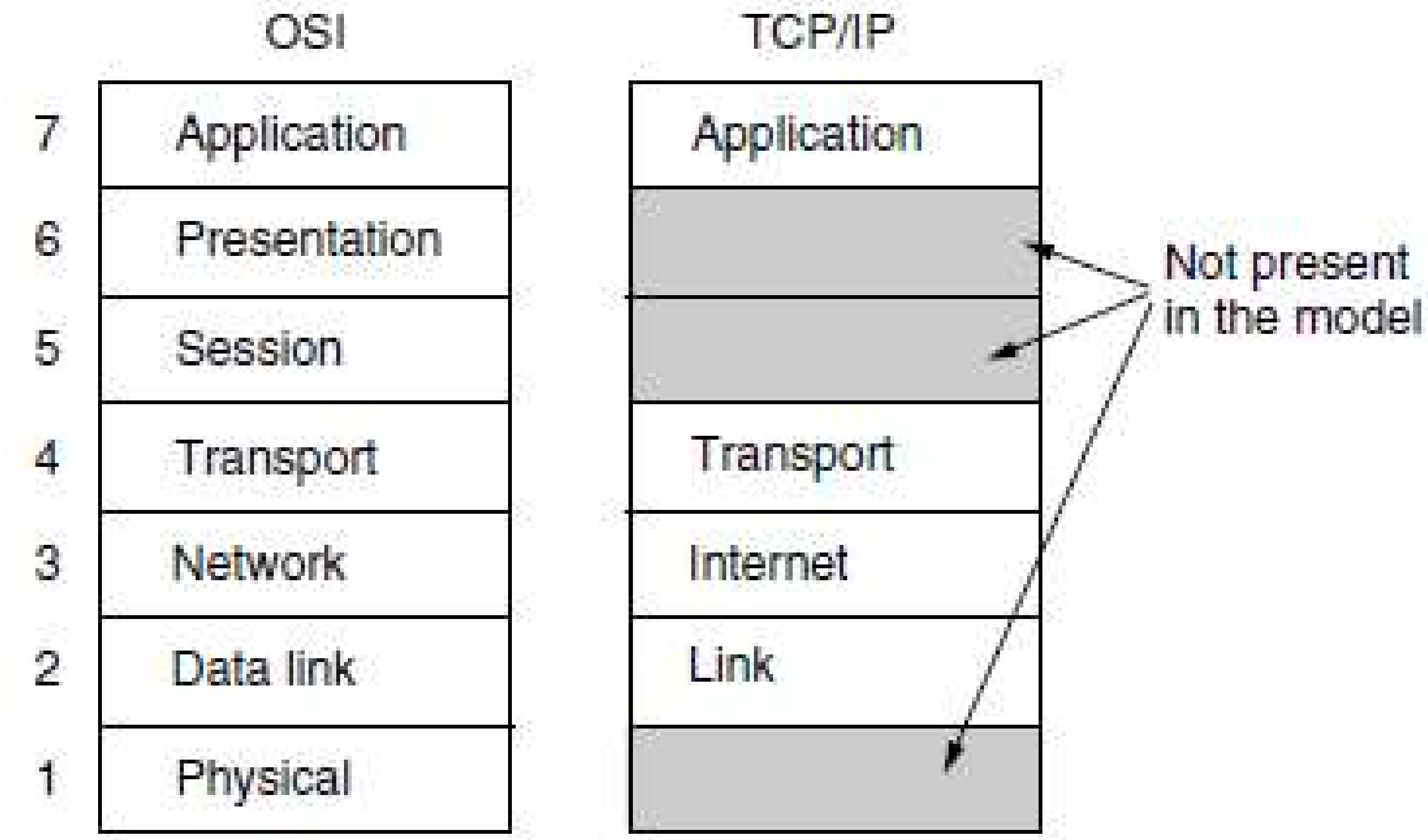
Presentation Layer(表示层)

- The syntax(语法) and semantics(语义) of the information transmitted
  - the data structures to be exchanged can be defined in an abstract way
  - Data encryption（数据加密）
  - Data compression （数据压缩）

Application Layer(应用层)

- Application layer contains a variety of protocols that are commonly needed by users.
  - WWW: HTTP (HyperText Transfer Protocol)
  - File transfer: FTP
  - Electronic mail: SMTP, POP3
  - BBS: Telnet
  - …….

The TCP/IP Reference Model



The Internet Layer(网际层)

- A packet-switching network based on a connectionless internetwork layer
- Defines an official packet format and protocol called IP (Internet Protocol)
- The job of the internet layer is to deliver IP packets where they are supposed to go
  - Packet routing
  - Avoiding congestion

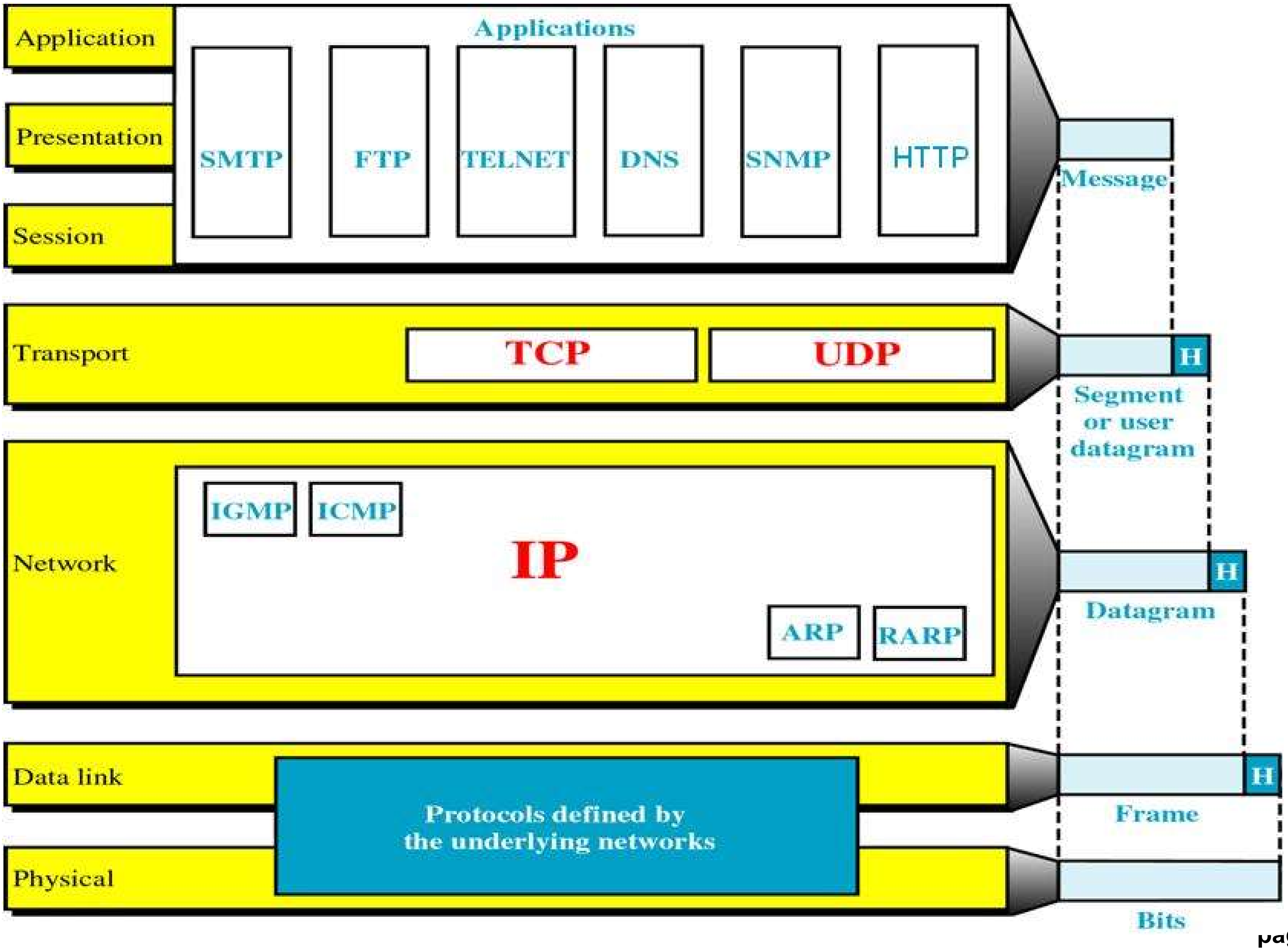
The Transport Layer: TCP

- TCP (Transmission Control Protocol)
  - Reliable connection-oriented
  - byte stream(字节流): Segments incoming byte stream into discrete messages
  - Reassembles the received messages into the output stream
  - Flow control(流量控制)
  - Congestion control(拥塞控制)

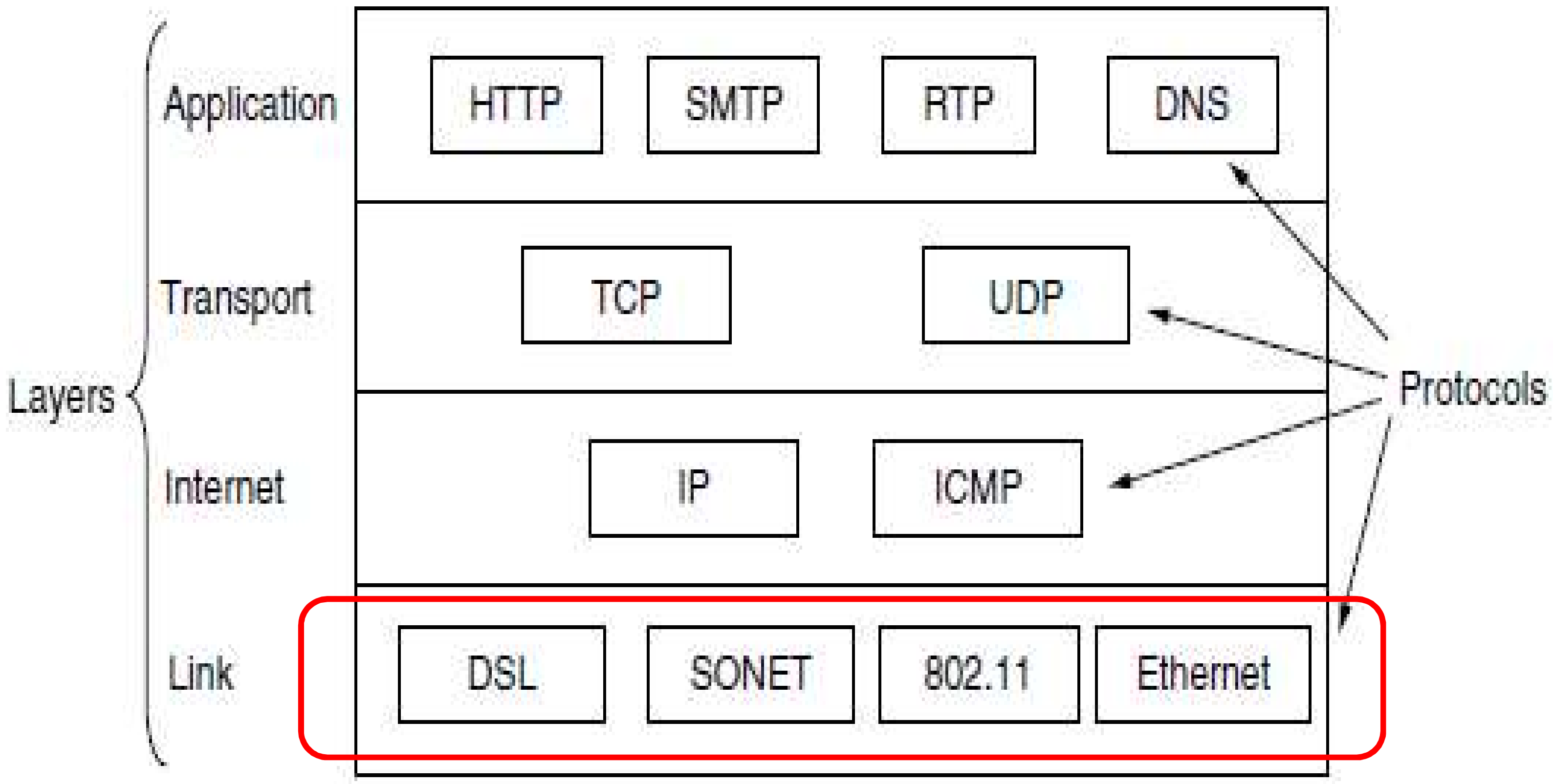
The Transport Layer: UDP

- UDP (User Datagram Protocol)
  - Unreliable, connectionless protocol
  - For applications that do not want TCP ‘s sequencing or flow control and wish to provide their own (i.e. NFS 网络文件系统)
  - For applications in which prompt delivery is more important than accurate delivery, such as transmitting speech or video, DNS

TCP/IP Model: in Details



Protocols & networks in the TCP/IP



OSI features: 3 key concepts

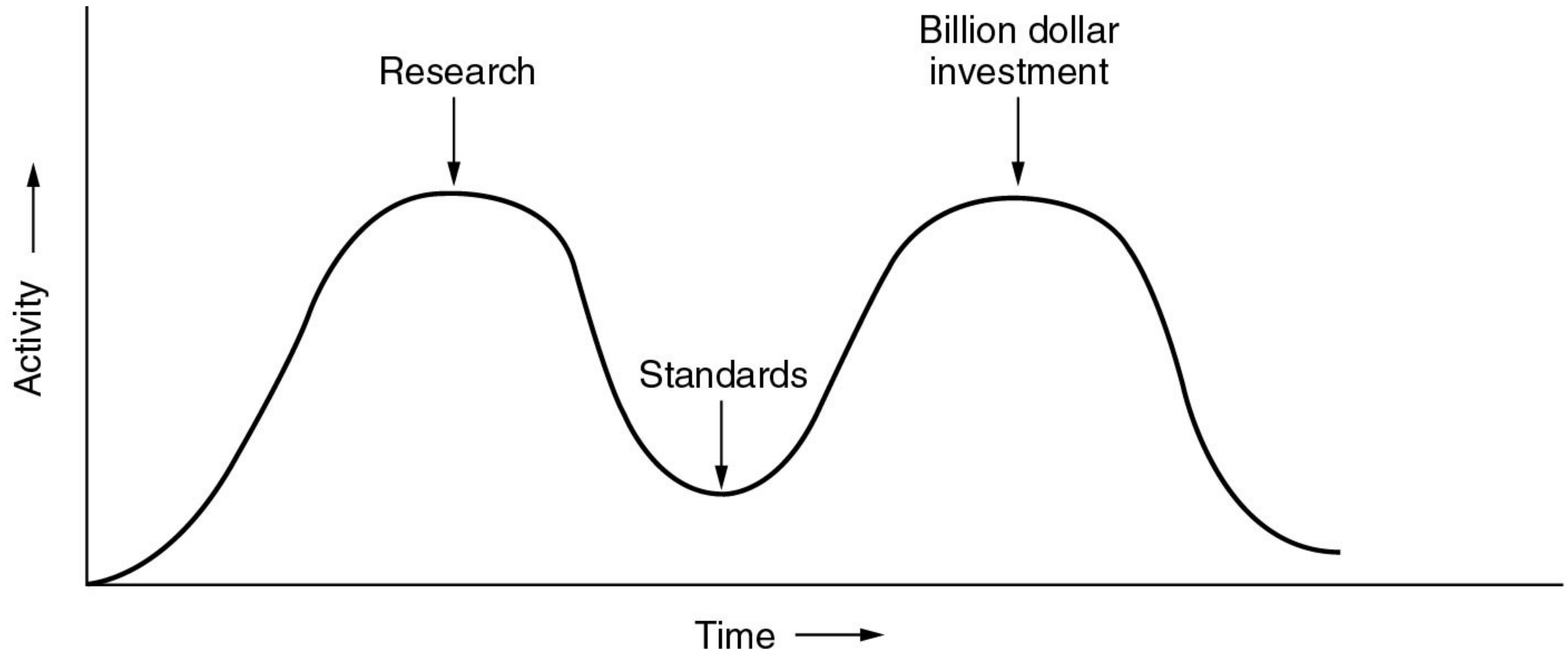
- Services (semantics)
  - What the layer does, not how entities above it access it or how the layer works.
  - 服务：某层实体对于上一层实体的支持
- Interfaces (syntax)
  - Tells the processes above it how to access it.
  - 接口：定义某层实体对于上一层提供的原语操作
- Protocols
  - 协议：两个系统同层的对等实体进行通信所必须遵守的规则

OSI: Critique

- Why OSI did not take over the world
  - Bad timing
  - Bad technology
  - Bad implementations
  - Bad politics

Bad Timing

- The apocalypse of the two elephants.



Bad Technology

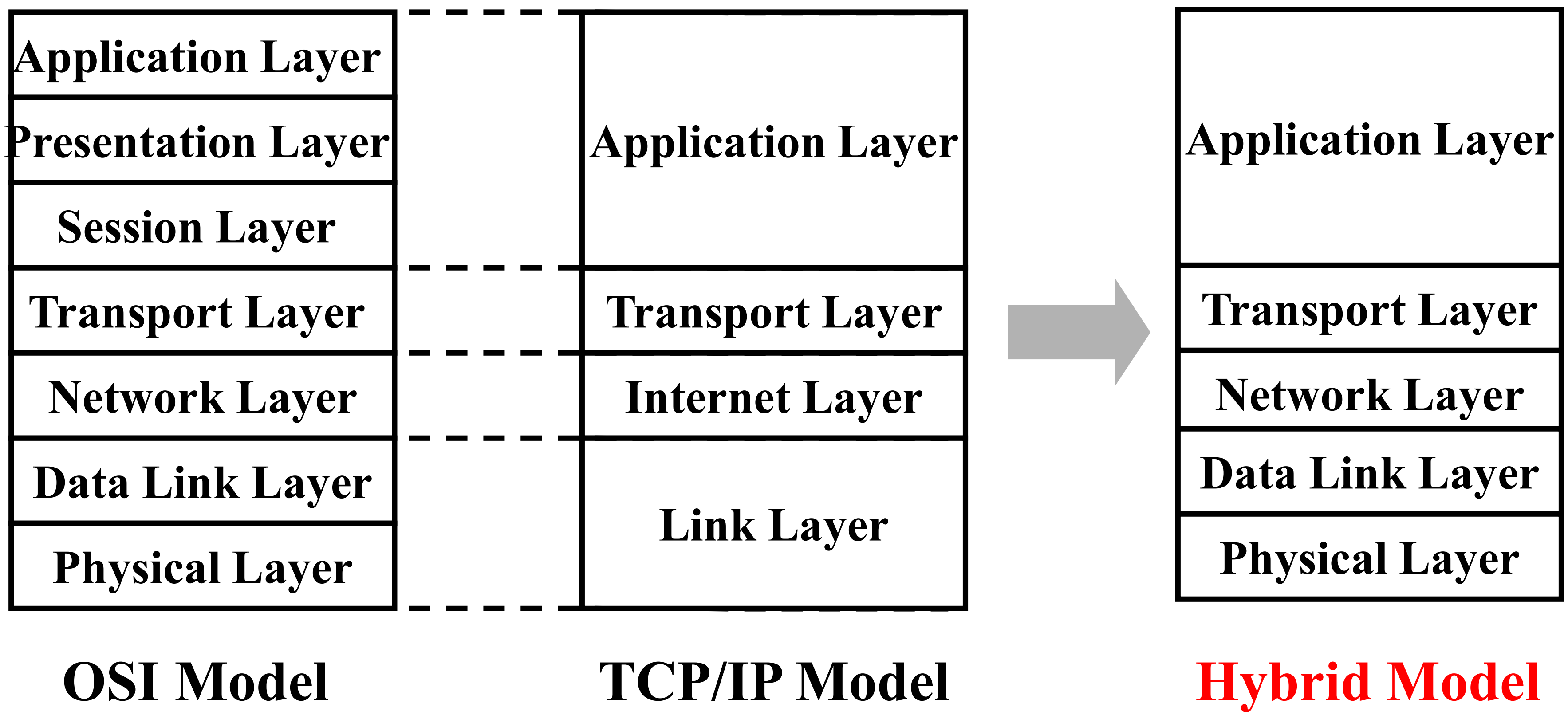
- The choice of 7 layers was more political than technical, and two of the layers (session and presentation) are nearly empty, whereas two other ones (data link and network) are overfull
- Service definitions and protocols are extraordinarily complex and difficult to implement and inefficient in operation
- Some functions, such as addressing, flow control, and error control, reappear again and again in each layer

TCP/IP: Critique

- Problems
  - Service, interface, and protocol not distinguished
  - Not a general model
  - "Link layer" not really a layer
  - No distinguishing between physical and data link layers
  - Minor protocols deeply entrenched, hard to replace, eg. Telnet

Hybrid Model

- The hybrid reference model used in this course



Key Points in Chapter 1

- What is a computer network?
  - Computer networks vs. communication networks
  - Computer networks vs. distributed systems
  - Host, node and terminal
  - Categories of computer networks
    - Broadcasting vs. point-to-point
    - LAN vs. WAN
- Network architecture
  - Layering, Information flow, Encapsulation
  - Protocol vs. service
  - Services: connection-oriented vs. connectionless
  - Peer, PDU
- Practical Layering Models(OSI RM vs. TCP/IP suite)
  - Number of layers and function of each layer
  - Good things and bad things

Homework

- Textbook, chapter 1, p105-108

Q1-17: 1, 2, 3, 4, 5, 9, 10, 11, 12, 15, 16, 17, 18, 20, 22, 30, 35

Metric Units

Exp.	Explicit	Prefix	Exp.	Explicit	Prefix
10^{-3}	0.001	milli	10^{3}	1,000	Kilo
10^{-6}	0.000001	micro	10^{6}	1,000,000	Mega
10^{-9}	0.000000001	nano	10^{9}	1,000,000,000	Giga
10^{-12}	0.000000000001	pico	10^{12}	1,000,000,000,000	Tera
10^{-15}	0.000000000000001	femto	10^{15}	1,000,000,000,000,000	Peta
10^{-18}	0.000000000000000001	atto	10^{18}	1,000,000,000,000,000,000	Exa
10^{-21}	0.000000000000000000001	zepto	10^{21}	1,000,000,000,000,000,000,000	Zetta
10^{-24}	0.000000000000000000000001	yocto	10^{24}	1,000,000,000,000,000,000,000,000	Yotta

The principal metric prefixes

milli and micro both begin with the letter ''m', Normally, ''m'' is for milli and ''μ'' (the Greek letter mu) is for micro

Terms

缩写	英文全称	中文
	access network	接入网
AP	Access Point	接入点
	acknowledged datagram	确认/应答数据报
	addressing	寻址
	ad hoc network	自组织网络
	application layer	应用层
	base station	基站
	bluetooth	蓝牙
	botnet	僵尸网络
	broadcasting	广播
	broadband	宽带
	cable headend	线缆头端
	cellular networks	蜂窝网络
	circuit	电路

缩写	英文全称	中文
	classic Ethernet	传统以太网
C/S	Client/Server	客户-服务器
CDMA	Code Division Multiple Access	码分多址
	communication subnet	通信子网
	connectionless	无连接
	connection-oriented	面向连接
	connectivity	连通性
	congestion	拥塞
	core network	核心网
	cut-through switching	直通交换
	data link layer	数据链路层
	datagram	数据报
	dial-up	拨号
DSL	Digital Subscriber Line	数字用户环线
DSLAM	Digital Subscriber Line Access Multiplexer	数字用户环线接入复用器

缩写	英文全称	中文
	distributed system	分布式系统
	e-commerce	电子商务
	encapsulation	封装
	error detection	差错检测
	Ethernet	以太网
	flow control	流量控制
	forwarding algorithm	转发算法
	frame	帧
FTTH	Fiber to the Home	光纤到户
	gateway	网关
GPS	Global Positioning System	全球定位系统
GSM	Global System for Mobile communications	全球移动通信系统
	host	主机
	hotspots	热点
	HUB	集线器

缩写	英文全称	中文
	interface	接口
	internet	互联网
	Internet	因特网
	internet layer	网际层
	internetworks	互联网
	internetworking	互通
	Interplanetary Internet	星际互联网
IoT	Internet of Things	物联网
IP	Internet Protocol	网际互连协议
	IP Telephony	IP电话
IPTV	IP Television	IP电视
IS	Instant Messaging	即时消息
ISP	Internet Service Provider	互联网服务提供商
LAN	Local Area Network	局域网
MAN	Metropolitan Area Network	城域网
	middleware	中间件
	mobile-commerce	移动商务

缩写	英文全称	中文
	multicasting	多播/组播
	negotiation	协商
	Network Architecture	网络体系结构
NFC	Near Field Communication	近场通信
	network service provider	网络服务提供商
	node	节点
OS	Operating System	操作系统
OSI	Open System Interconnection	开放系统互连
PAD	Personal Digital Assistant	个人数字助手(平板电脑)
PAN	Personal Area Network	个域网
	package	分组/包
PDU	Packet Data Unit	协议数据单元
	phishing messages	钓鱼消息
	physical layer	物理层
	physical medium	物理媒体/介质
	Port	端口
	presentation layer	表示层

缩写	英文全称	中文
	protocol	协议
	protocol layering	协议分层
	protocol stack	协议栈
PLC	Power Line Communications	电力线通信
	power-line networks	电力线网络
PSTN	Public Switched Telephone Network	公众交换电话网络
	peer	对等（实体）
	peer-to-peer	对等（网络）
QoS	Quality of Service	服务质量
	Reference Model	参考模型
RFC	Request For Comment	请求评论(IETF的标准)
	request-reply	请求-应答
RFID	Radio Frequency Identification	射频识别
	router	路由器
	routing algorithm	选路算法
	scalable	可扩展的

缩写	英文全称	中文
	service	服务
SAP	Service Access Point	服务访问点
SMS	Short Message Service	短消息业务
	sensor networks	传感网
	Service Primitives	服务原语
	session layer	会话层
	social network	社区网
	spam	垃圾邮件
	statistical multiplexing	统计复用
	store-and-forward switching	存储转发交换
	subnet	子网
	switch	交换机
	switched Ethernet	交换式以太网
TCP	Transmission Control Protocol	传输控制协议
	terminal	终端
	token management	令牌管理
	transmission lines	传输线路

Terms(8)

缩写	英文全称	中文
	transport layer	传输层
	ubiquitous computing	普适计算
UDP	User Datagram Protocol	用户数据报协议
	Unicasting	单播
VLAN	Virtual LAN	虚拟局域网
VoIP	Voice over IP	IP话音
VPN	Virtual Private Networks	虚拟专用网
WAN	Wide Area Network	广域网
WCDMA	Wideband Code Division Multiple Access	宽带码分多址接入
	wearable computers	可穿戴计算机
WLAN	Wireless Local Area Network	无线局域网
WWW	World Wide Network	万维网

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- Some figures used in this presentation have been either directly copied or adapted from following materials
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  - “Data and Computer Communications”, which are marked with *[Stallings]*
  - “计算机网络”, 谢希仁，第五版，电子工业出版社，2008年，which are marked with *[谢]*

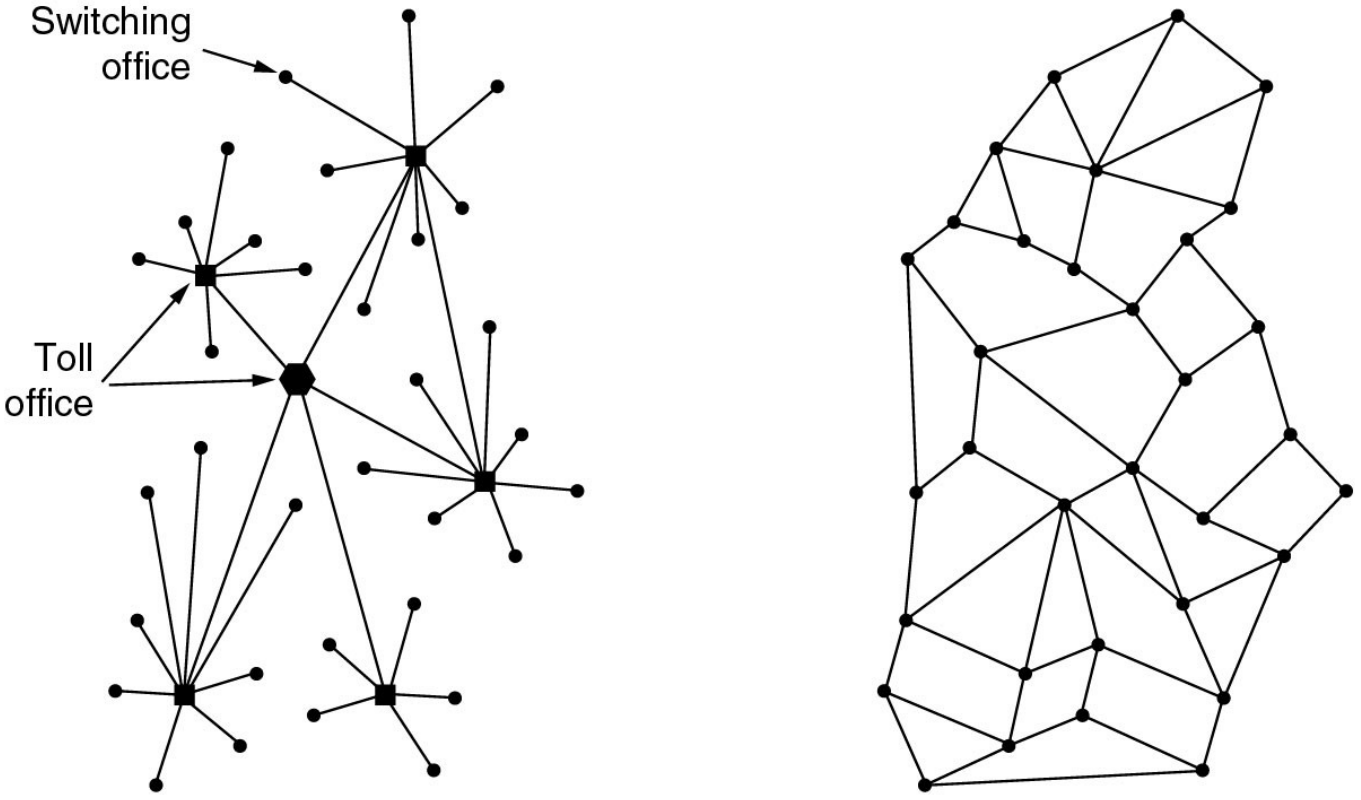
Self-study after class

Outline

- What is a Computer Network?
- What can we do with Computer Networks?
- Categories of Computer Networks
- Network architecture and protocols
- Reference Models
- *Example Networks*
  - The Internet
  - Connection-Oriented Networks: X.25, Frame Relay, and ATM
  - Ethernet
  - Wireless LANs: 802:11
- Network Standardization

The ARPANET

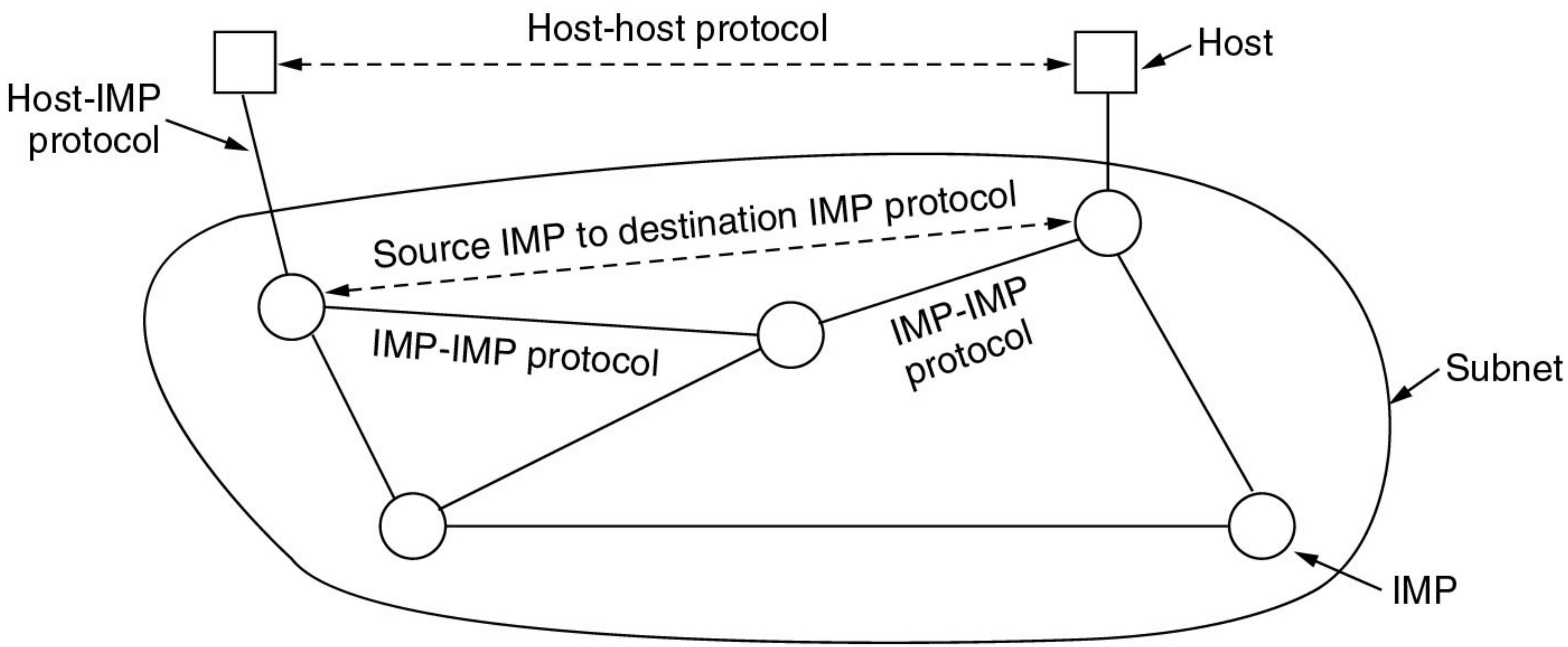
(a) Structure of the telephone system
(b) Baran's proposed distributed switching system



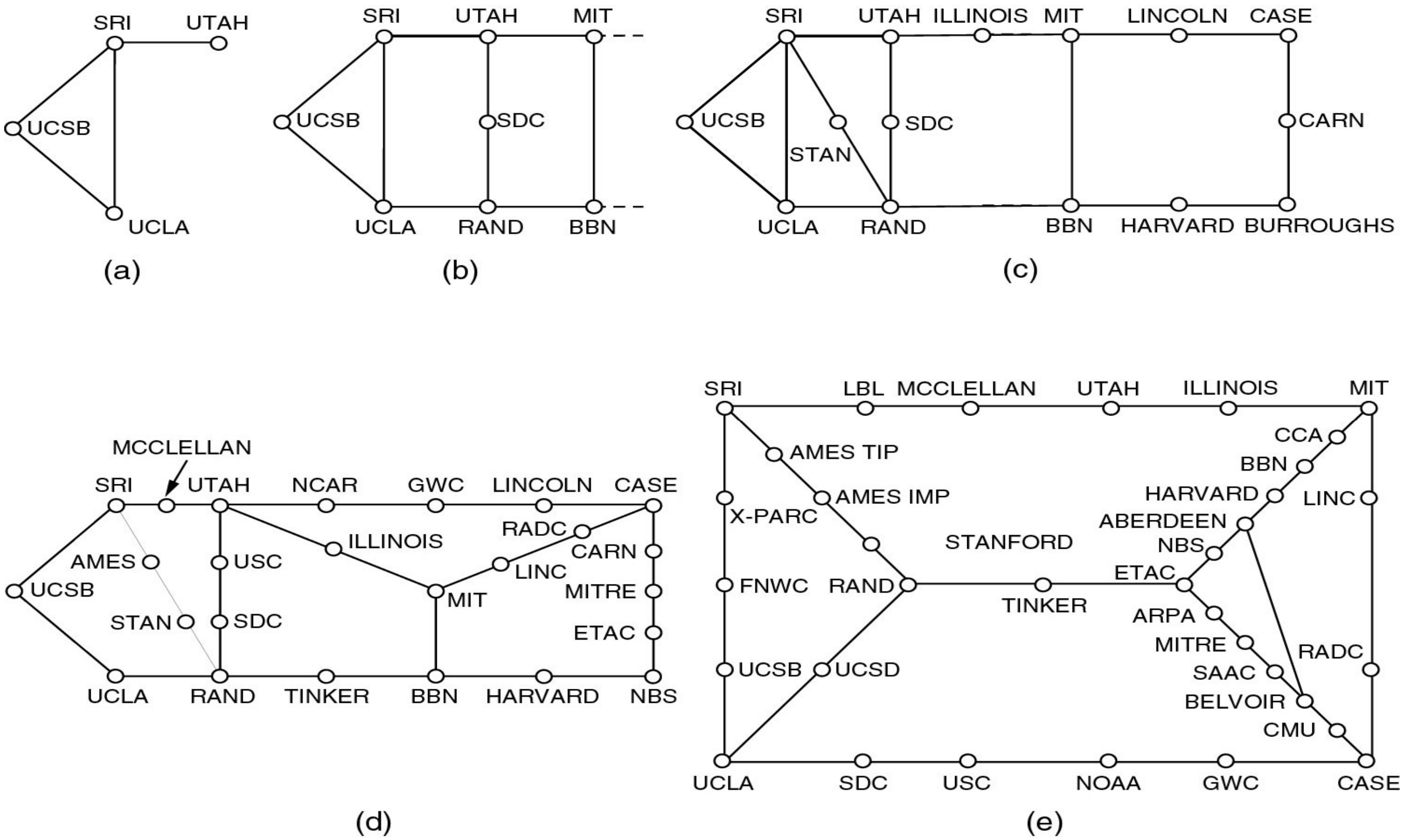
(a)

(b)

The Original ARPANET Design



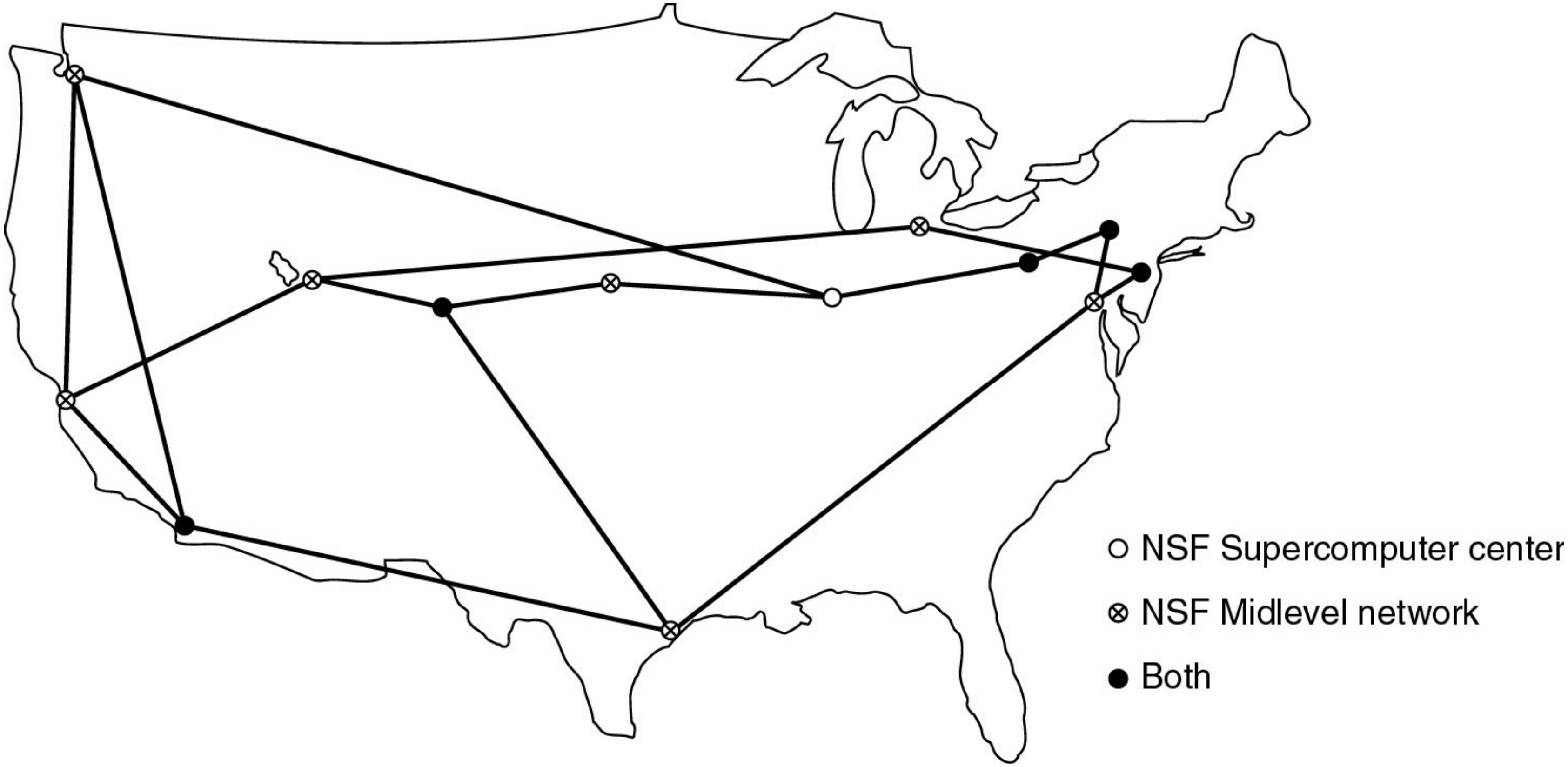
Growth of the ARPANET



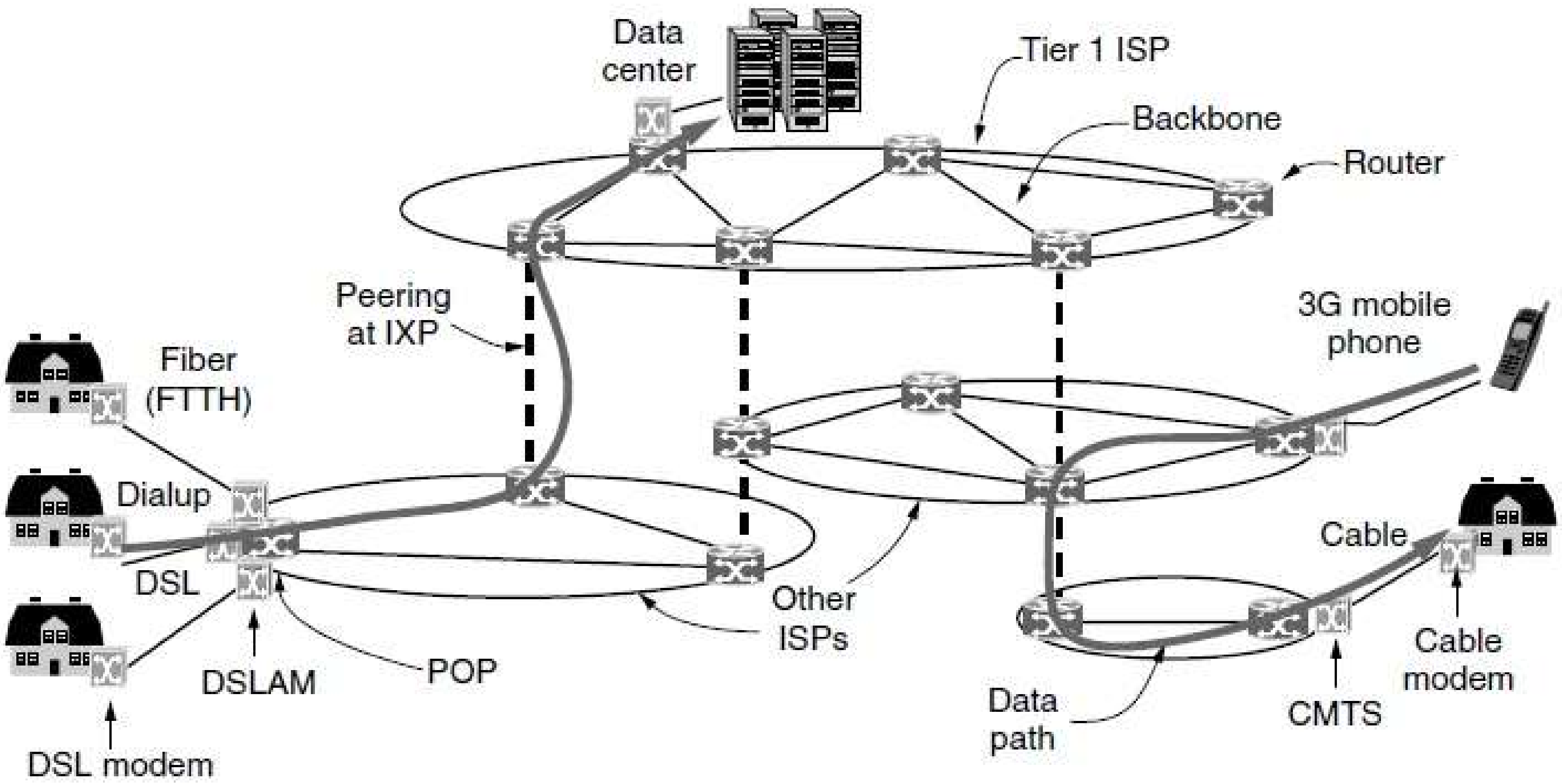
(a) December 1969 (b) July 1970 (c) March 1971
(d) April 1972. (e) September 1972.

NSFNET

- The NSFNET backbone in 1988.



Internet Architecture: Overview



POP: Point of Presence
IXP: Internet eXchange Point
DSL: Digital Subscriber Line
DSLAM: DSL Access Multiplexer
CMTS: Cable Modem Termination System

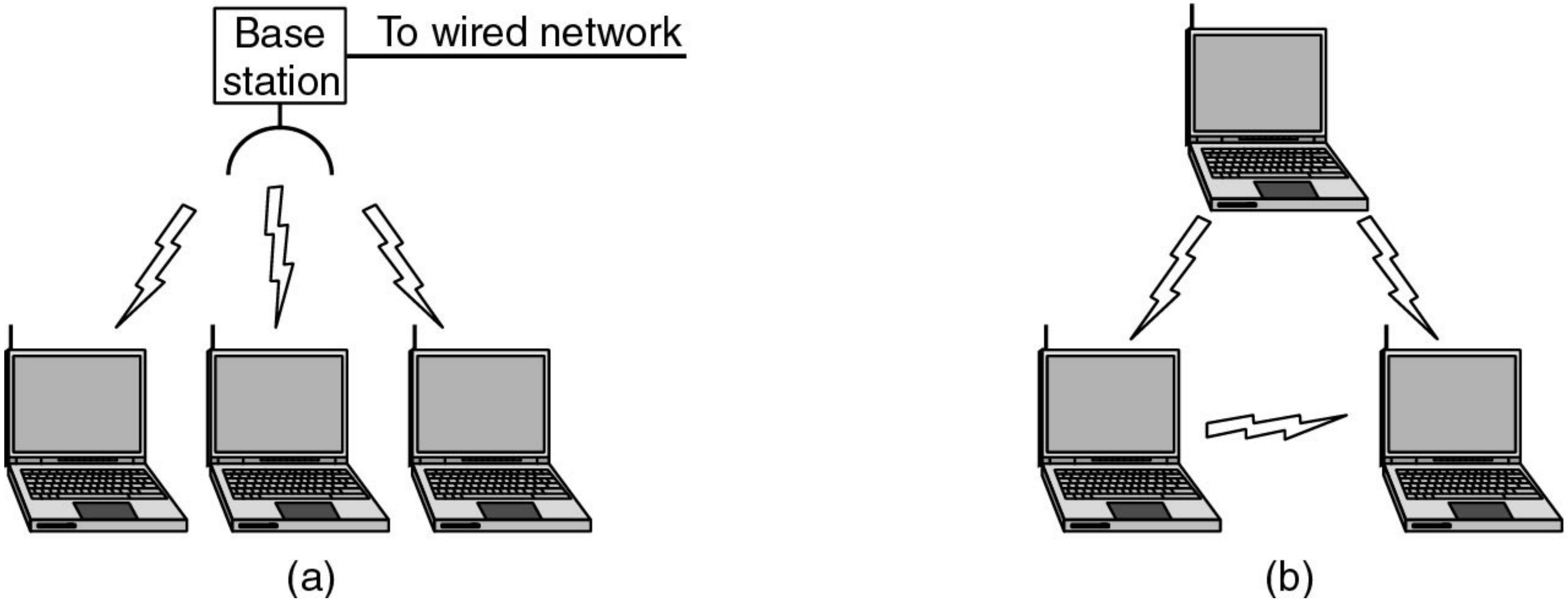
Internet Applications

- Traditional applications (1970 – 1990)
  - E-mail
  - News
  - Remote login
  - File transfer
- 1990-
  - WWW (World Wide Web)
  - Instant messaging
  - P2P: BitTorrent
  - Multimedia applications
    - IP Telephony, Streaming media, IPTV
  - Blog, Weibo, Weixin

Other Networks

- 3G，4G Moblie Phone Networks
- Wireless LAN
- RFID and Sensor networks

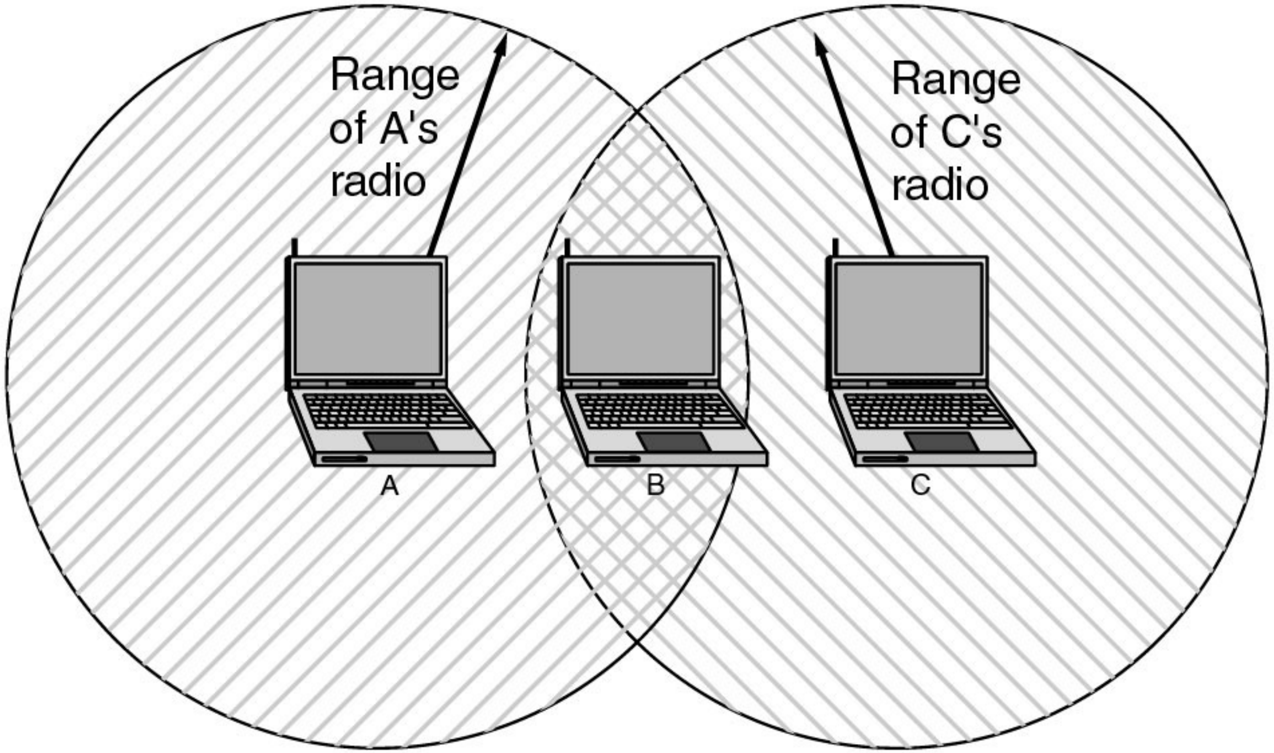
Wireless LANs



(a) Wireless networking with a base station.
(b) Ad hoc networking.

Wireless LANs (2)

- The range of a single radio may not cover the entire system.



Outline

- What is a Computer Network?
- What can we do with Computer Networks?
- Categories of Computer Networks
- Network architecture and protocols
- Reference Models
- Example Networks
- *Network Standardization*

Network Standardization

- Telecommunications World
  - ITU (International Telecommunication Union)
  - Main sectors
    - Radio communications(ITU-R)
    - Telecommunications Standardization(ITU-T)
      CCITT (1956-1993)
    - Development(ITU-D)
- International Standards World
  - ISO (International Standards Organization)
    - ANSI (American National Standards Institute)
  - NIST (National Institute of Standards and Technology)
  - IEEE (Institute of Electrical and Electronics Engineers)

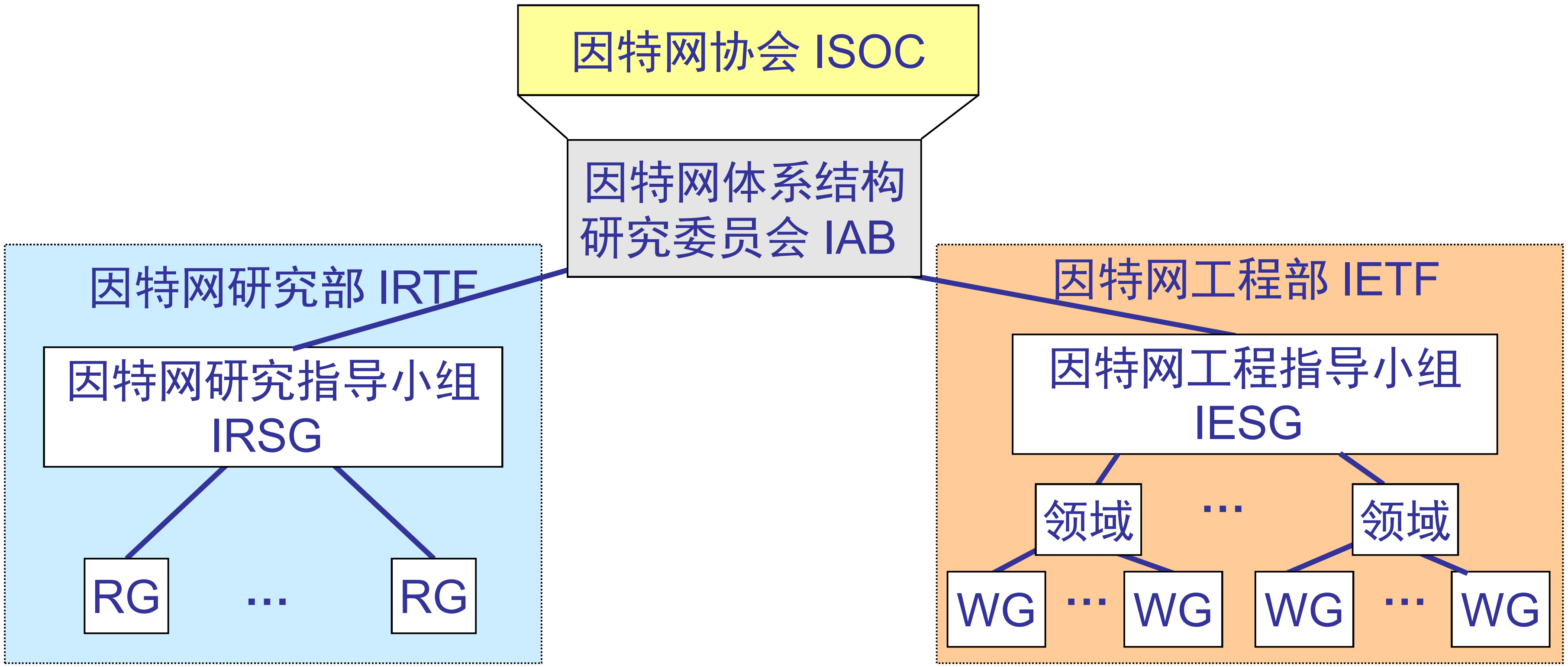
IEEE 802 Standards

Number	Topic
802.1	Overview and architecture of LANs
802.2 ↓	Logical link control
802.3 *	Ethernet
802.4 ↓	Token bus (was briefly used in manufacturing plants)
802.5	Token ring (IBM's entry into the LAN world)
802.6 ↓	Dual queue dual bus (early metropolitan area network)
802.7 ↓	Technical advisory group on broadband technologies
802.8 †	Technical advisory group on fiber optic technologies
802.9 ↓	Isochronous LANs (for real-time applications)
802.10 ↓	Virtual LANs and security
802.11 *	Wireless LANs (WiFi)
802.12 ↓	Demand priority (Hewlett-Packard's AnyLAN)
802.13	Unlucky number; nobody wanted it
802.14 ↓	Cable modems (defunct: an industry consortium got there first)
802.15 *	Personal area networks (Bluetooth, Zigbee)
802.16 *	Broadband wireless (WiMAX)
802.17	Resilient packet ring
802.18	Technical advisory group on radio regulatory issues
802.19	Technical advisory group on coexistence of all these standards
802.20	Mobile broadband wireless (similar to 802.16e)
802.21	Media independent handoff (for roaming over technologies)
802.22	Wireless regional area network

Internet Standards World(1)

- Internet协会（ISOC，Internet Society）
  - 推动、支持和促进Internet不断增长和发展的专业组织
  - 它把Internet作为全球研究通信的基础设施
- 体系结构委员会（IAB，Internet Architecture Board）
  - 技术监督和协调的机构。它由国际上来自不同专业的15个志愿者组成，其职能是负责Internet标准的最后编辑和技术审核
- 工程任务组（IETF，Internet Engineering Task Force）
  - 面向近期标准的组织，它分为9个领域（应用、寻径和寻址、安全等等）。IETF开发成为Internet标准的规范
- 研究任务组（IRTF，Internet Research Task Force）
  - 主要对长远的项目进行研究
- IRTF和IETF隶属于IAB，IAB隶属于ISOC
  - 所有关于Internet的标准都以**RFC(Request for Comments)**文档出版
  - Proposed Standard->Draft Standard->Internet Standard

Internet Standards World(2)



- RG: Research Group
- WG: Working Group

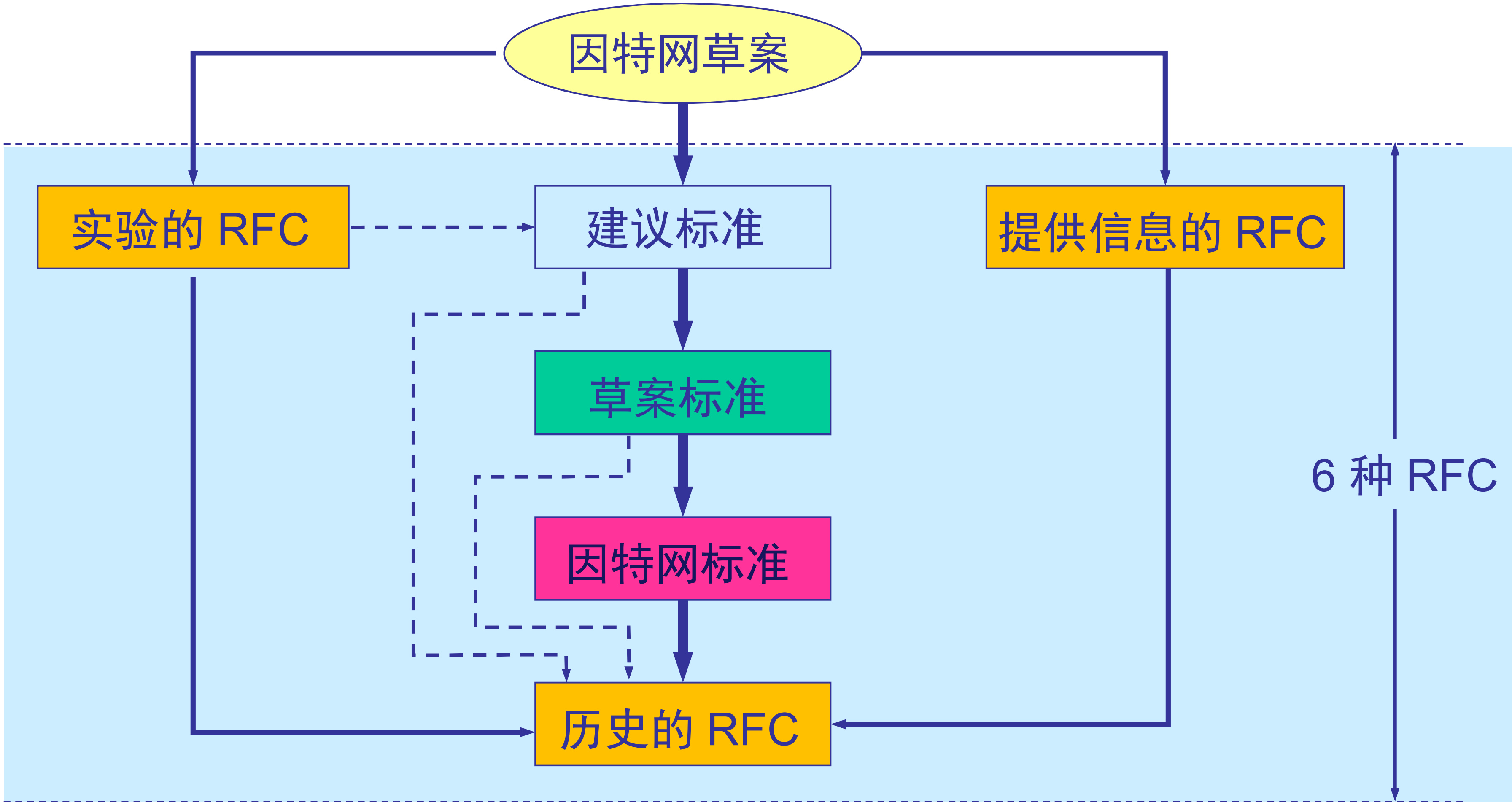
[谢]

Standardization process(1)

- Internet Draft(因特网草案) ——在这个阶段还不是 RFC 文档。
- Proposed Standard(建议标准) ——从这个阶段开始就成为 RFC 文档。
- Draft Standard (草案标准)
- Internet Standard (因特网标准)

[谢]

Standardization process(2)



[谢]